

HCDE 511

Information Visualization

Winter 2026

2
Week



2 Week

Exercise: Visual Mappings

Assignment 1 share out & critique

Break

Graphical excellence & integrity

Critiquing visualizations

Break

Intro to Assignment 2

Project groups

Project proposal brainstorm

Week 1

Intro
Data
encoding &
mapping

2

Graphical
excellence
& integrity
Critiquing

3

Fundamen-
tal graphs
Working
with data

4

Visual
perception
& design

5

Tasks &
inter-
actions

6

Collabora-
tive
critique

7

Multi-
variate
analysis

8

Trees &
networks

9

Animation,
context,
story-
telling

10

Final pre-
sentations

Finals

A1 Good/bad

A2 Static vis

A3 Exploratory analysis

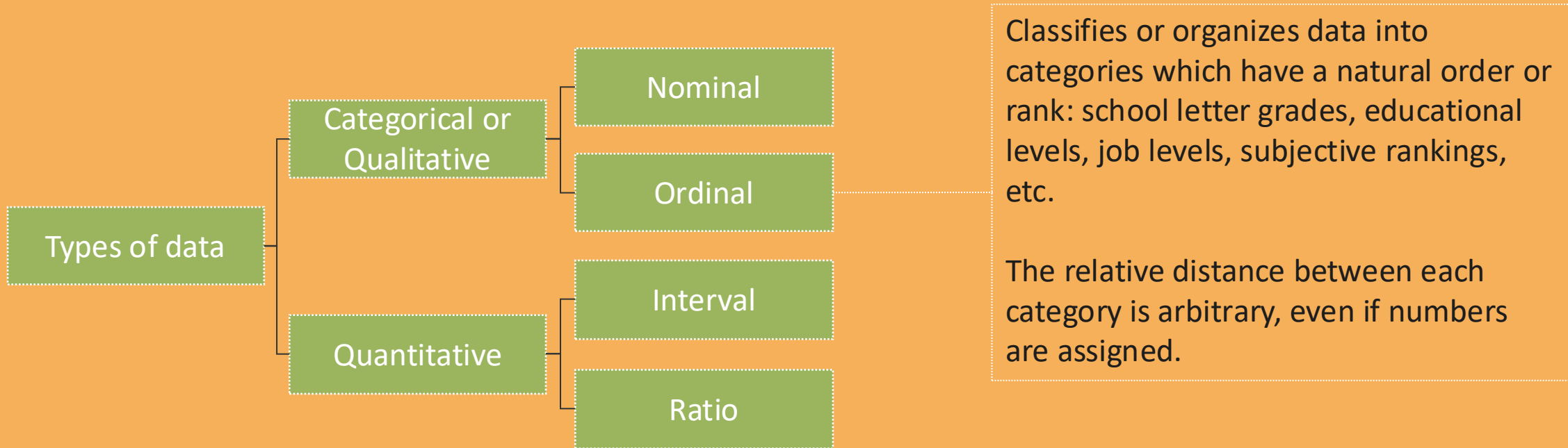
Project proposal

Rev. prop.

Final project deliverable

Final project writeup

	Qualitative or Categorical		Quantitative	
	Nominal	Ordinal	Interval	Ratio
Categories, labels, names	✓			
Rank/order categories, without inherent #'s		✓		
Has known equal intervals			✓	
Has true or meaningful zero				✓



Visual Mapping

In groups of 2:

- 1 List each data dimension (variable) and its data type (Nom, Ord, Quant)
- 2 Start with either the quantitative data or the data you want to emphasize
- 3 Select an effective visual encoding for this data variable based on effectiveness ranking
- 4 Check expressiveness of each visual encoding
- 5 Check consistency of each visual encoding
- 6 Repeat until you have encoded all data variables
- 7 Reorder (if necessary)
- 8 Capture your work and submit to Canvas

Expressiveness

A set of facts is **expressible** in a visual language if the sentences (i.e. the visualizations) in the language express all the facts in the set of data, and only the facts in the data.

Effectiveness

A visualization is more **effective** than another visualization if the information conveyed by one visualization is more readily perceived than the information in the other visualization.

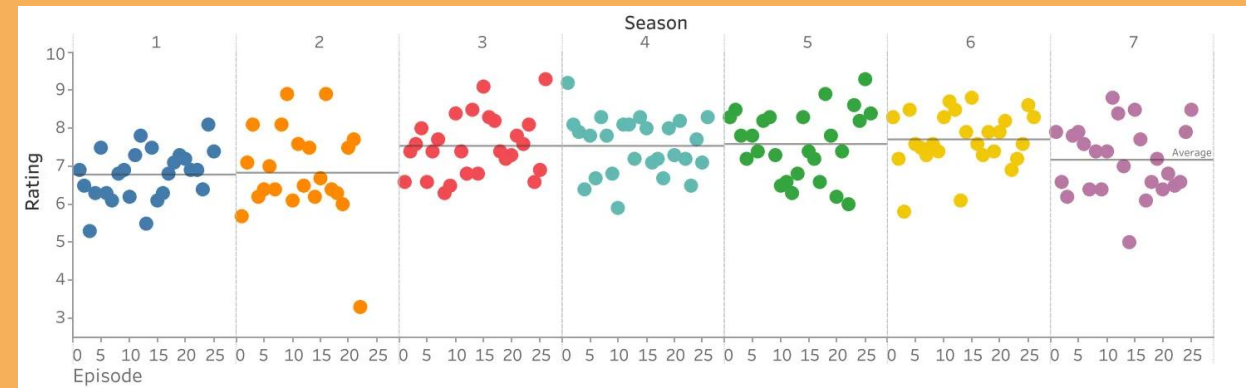
Visual Mapping

Example	Encoding	Ordered	Useful values	Quantitative	Ordinal	Categorical	Relational
	position, placement	yes	infinite	Good	Good	Good	Good
1, 2, 3; A, B, C	text labels	optional (alphabetical or numbered)	infinite	Good	Good	Good	Good
	length	yes	many	Good	Good		
	size, area	yes	many	Good	Good		
	angle	yes	medium/few	Good	Good		
	pattern density	yes	few	Good	Good		
	weight, boldness	yes	few		Good		
	saturation, brightness	yes	few		Good		
	color	no	few (< 20)			Good	
	shape, icon	no	medium			Good	
	pattern texture	no	medium			Good	
	enclosure, connection	no	infinite			Good	Good
	line pattern	no	few				Good
	line endings	no	few				Good
	line weight	yes	few		Good		

Noah Ilnsky

Guidance

Data	Data Type	Encoding	Note
Episode	Categorical	Position	Each episode is represented by a dot. Each dot has its own position on the canvas.
Episode Number	Ordinal	Position	The x-axis shows the number of each episode in each season.
Season	Ordinal	Color Position	Each season is represented by a different color (hue). Each season also has its own section on the chart.
Rating	Ordinal	Position	The better the episode, the higher it is on the y-axis.
Average season rating	Quantitative	Position	The horizontal bar in each pane shows the average rating of the episodes in each season. There is some controversy over whether you should average ordinal ratings. We believe that the practice is so common with ratings it is acceptable.



Wexler, et al.

Example

Assignment 1

Share Out

20 min

dinner break

until x:xx pm

Graphical
Excellence

Carte Figurative des pertes successives en hommes de l'Armée Française dans la campagne de Russie 1812-1813.

Dressée par M. Minard, Inspecteur Général des Ponts et Chaussées en retraite. Paris, le 20 Novembre 1869.

Les nombres d'hommes présents sont représentés par les largeurs des zones colorées à raison d'un millimètre pour dix mille hommes; ils sont de plus écrits en travers des zones. Le rouge désigne les hommes qui entrent en Russie, le noir ceux qui en sortent. Les renseignements qui ont servi à dresser la carte ont été puisés dans les ouvrages de M. M. Chiers, de Ségur, de Fezensac, de Chambray et le journal inédit de Jacob, pharmacien de l'Armée depuis le 28 Octobre.

Pour mieux faire juger à l'œil la diminution de l'armée, j'ai supposé que les corps du Prince Jérôme et du Maréchal Davoust qui avaient été détachés sur Minsk et Mohilow et ont rejoint vers Orscha et Witebsk, avaient toujours marché avec l'armée.

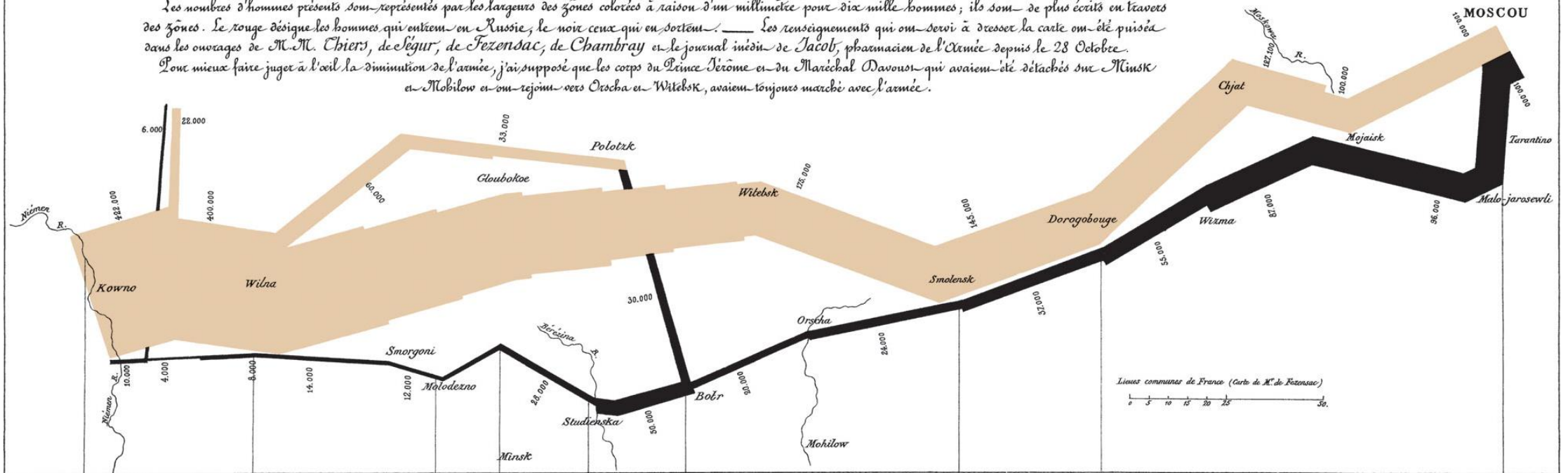
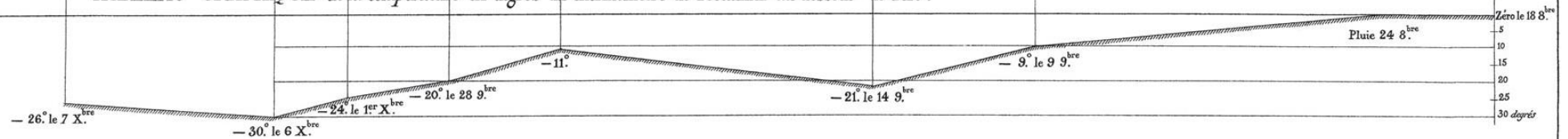


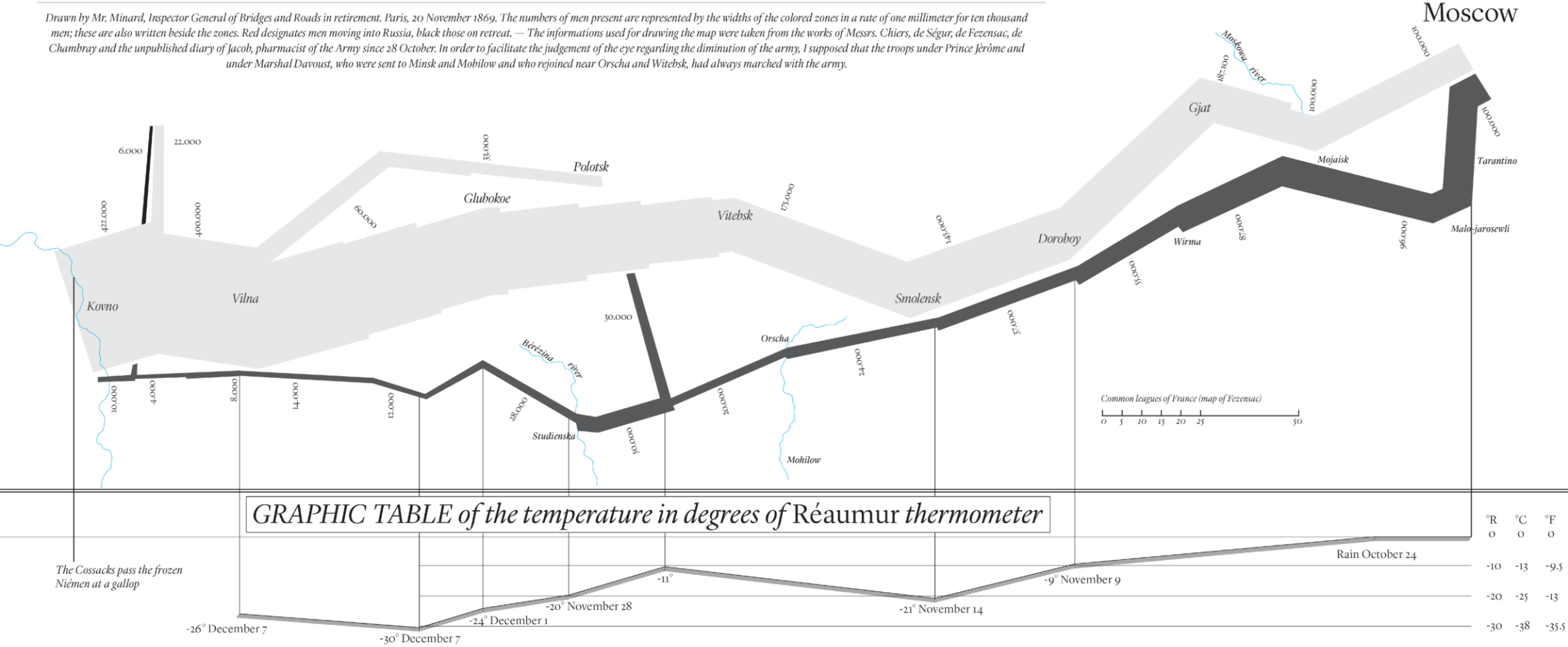
TABLEAU GRAPHIQUE de la température en degrés du thermomètre de Réaumur au dessous de zéro.

Les Cosaques passent au galop le Niémen gelé.

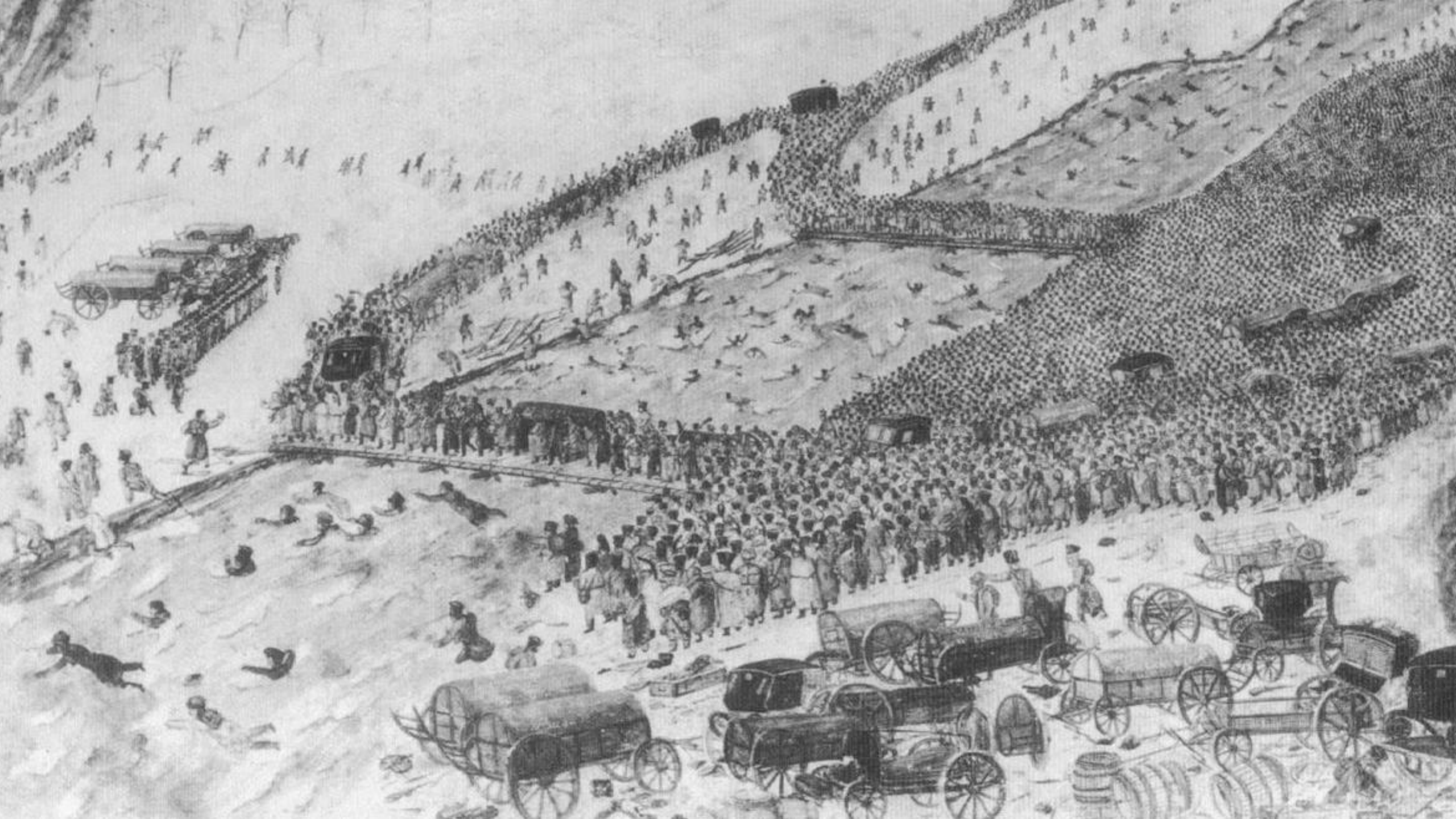


FIGURATIVE MAP of the successive losses in men of the French Army in the RUSSIAN CAMPAIGN OF 1812-1813

Drawn by Mr. Minard, Inspector General of Bridges and Roads in retirement. Paris, 20 November 1869. The numbers of men present are represented by the widths of the colored zones in a rate of one millimeter for ten thousand men; these are also written beside the zones. Red designates men moving into Russia, black those on retreat. — The informations used for drawing the map were taken from the works of Messrs. Chiers, de Ségur, de Fezensac, de Chambray and the unpublished diary of Jacob, pharmacist of the Army since 28 October. In order to facilitate the judgement of the eye regarding the diminution of the army, I supposed that the troops under Prince Jérôme and under Marshal Davoust, who were sent to Minsk and Mobilow and who rejoined near Orscha and Witebsk, had always marched with the army.





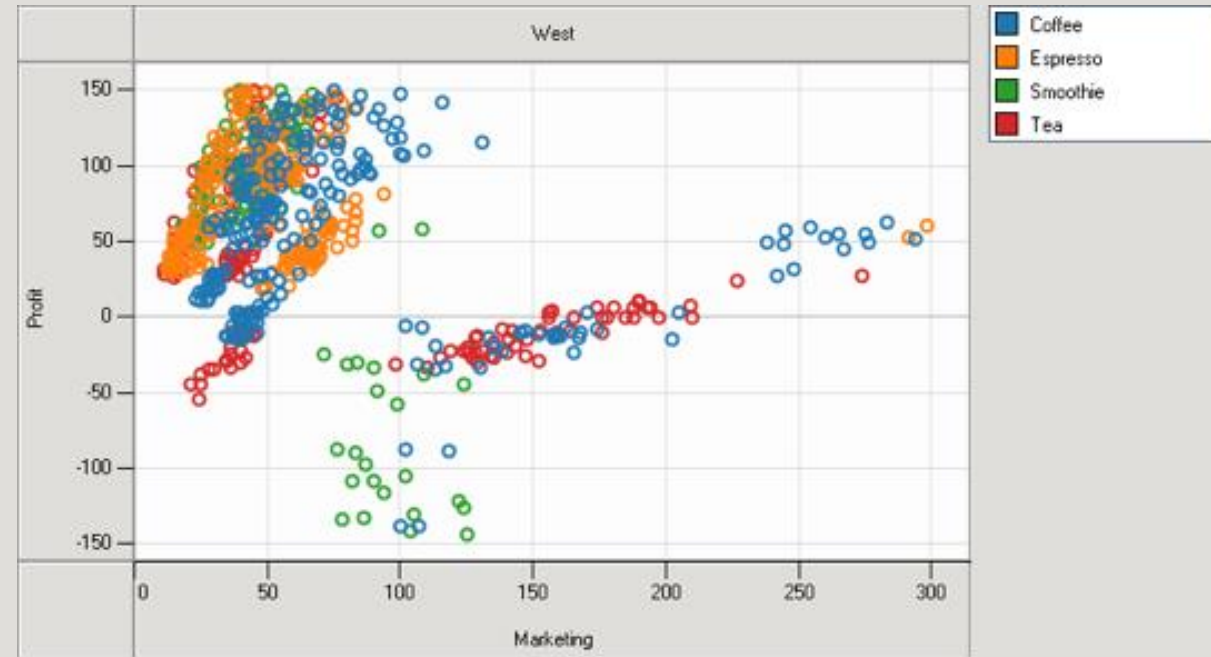


Greatest number of ideas
In the shortest time
With least “ink”
In the smallest space

Multiple variables:
Marketing, profit,
product type

Relationship between
variables

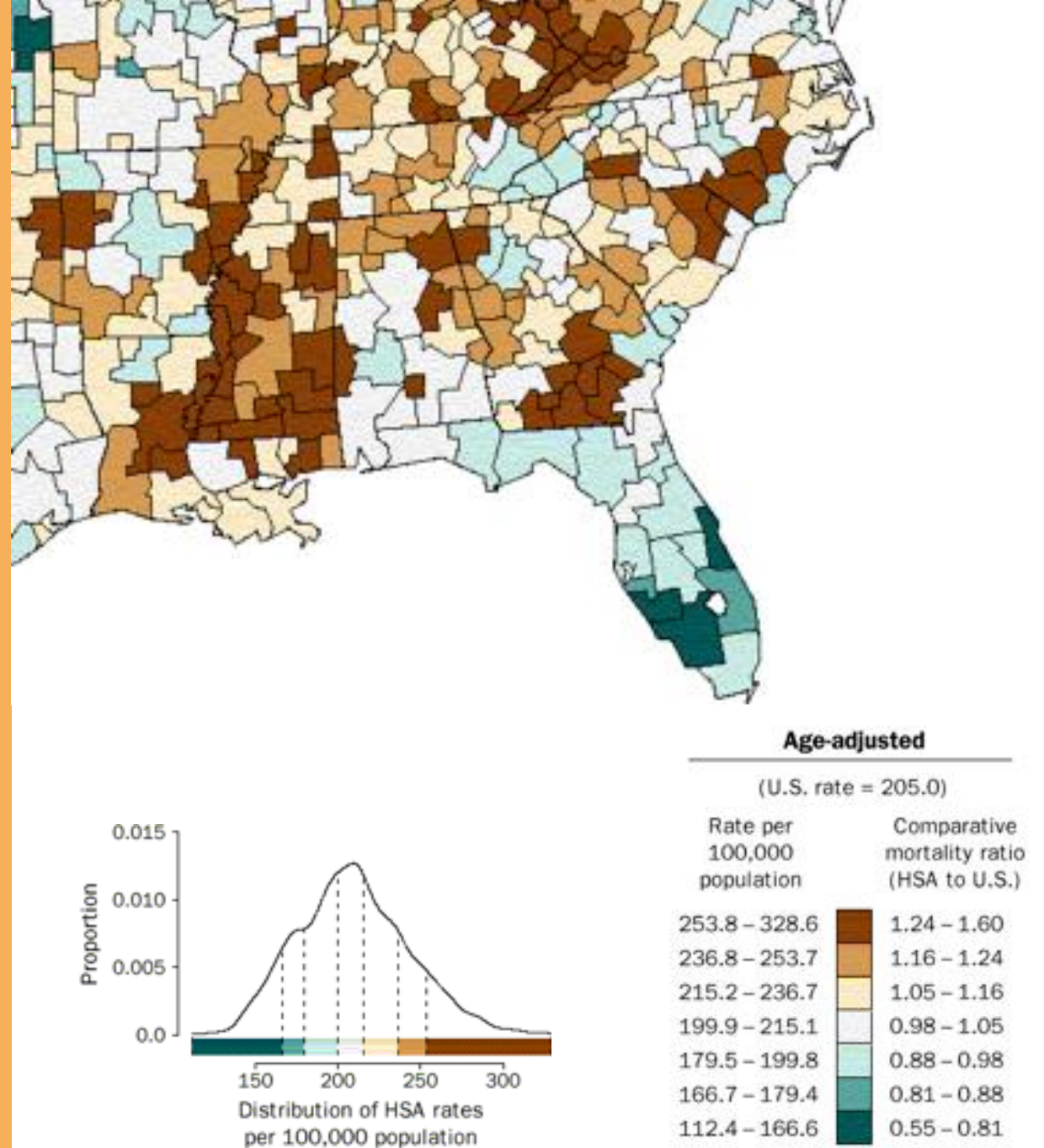
100s of values



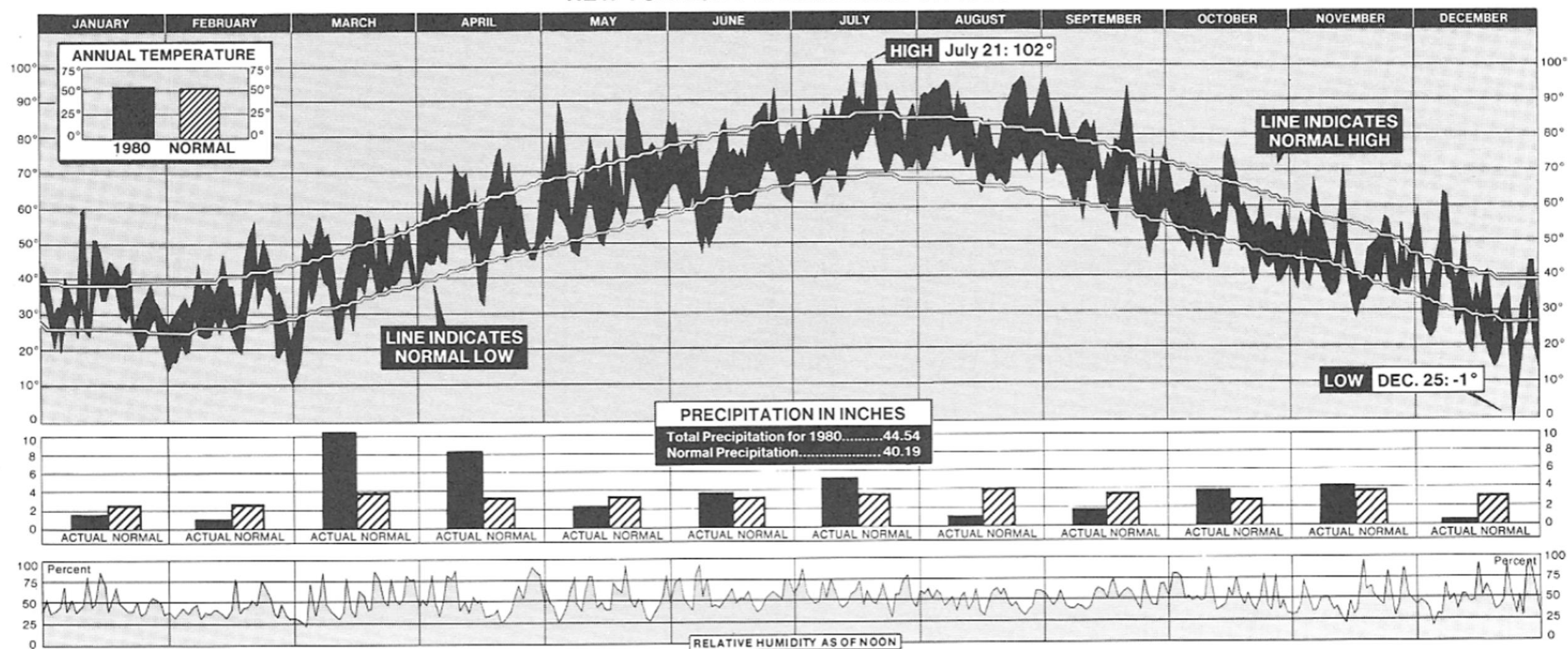
Multiple variables:
Location, mortality
ratio

Relationship between
variables

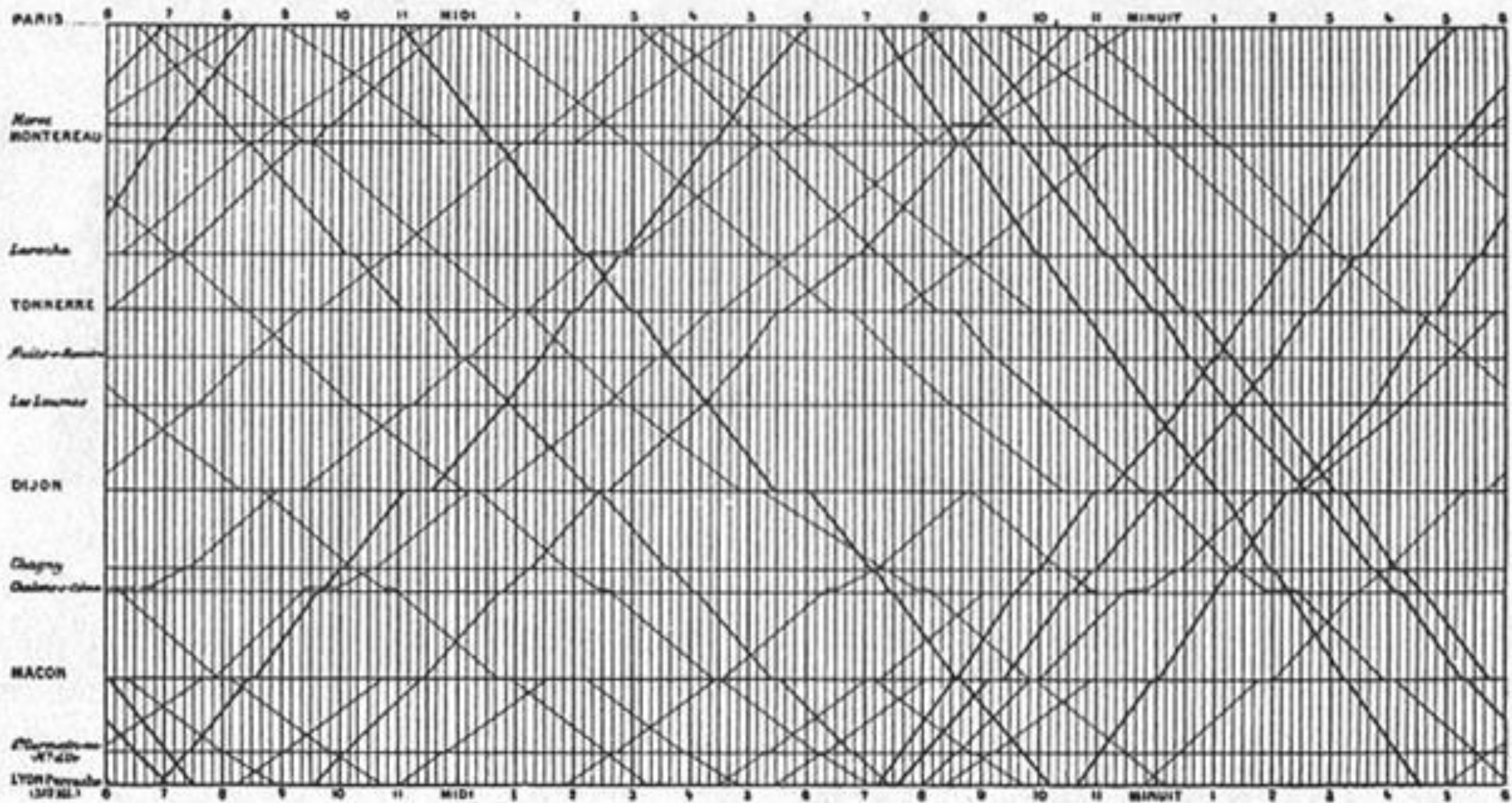
100,000s of values

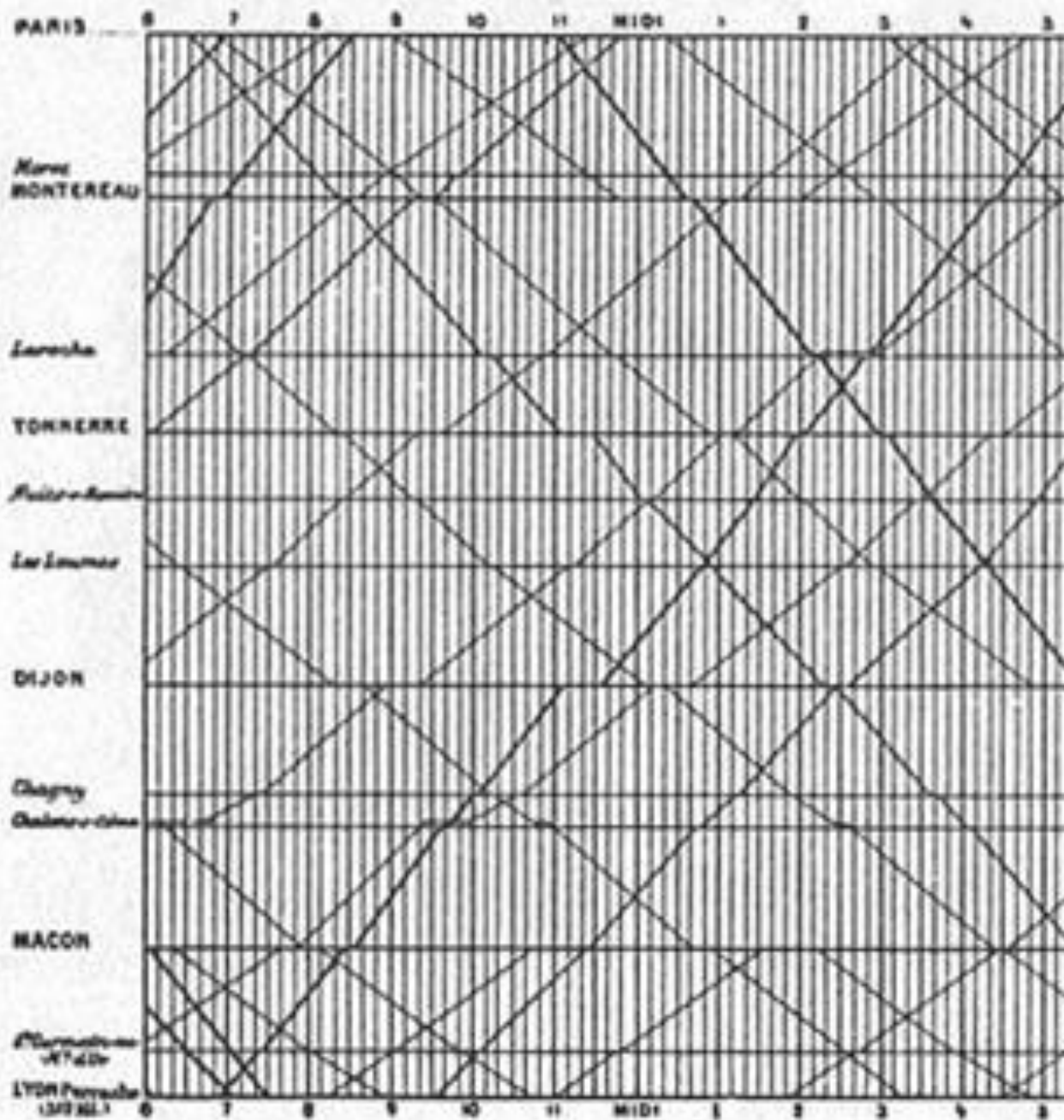


NEW YORK CITY'S WEATHER FOR 1980



New York Times, January 11, 1981, p. 32.





Arrival & departure times

Time at stop

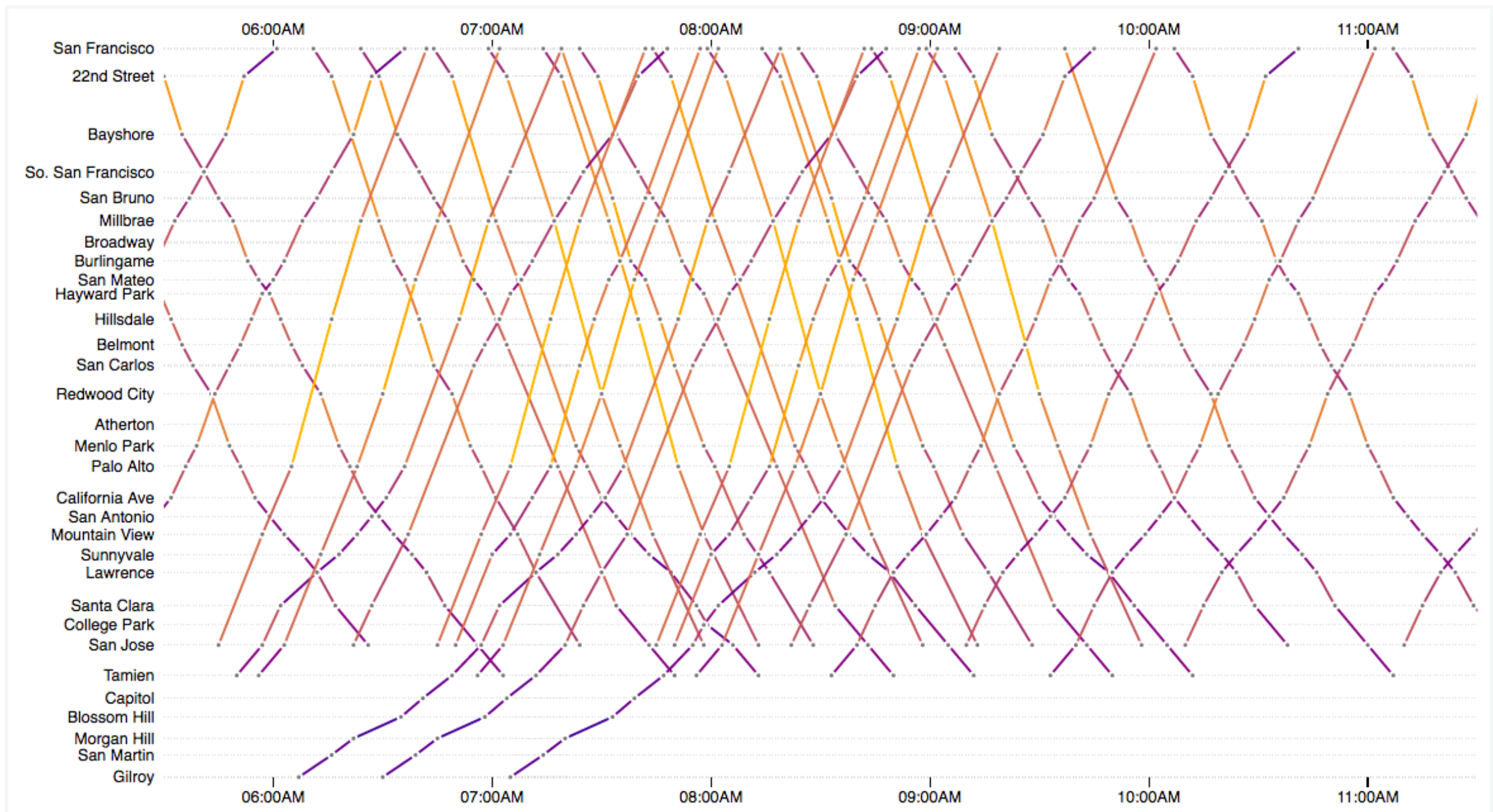
Number of trains

Direction

Relative speeds

Name & number of stops

Crossings



Show the data

Induce the viewer to think about the substance, rather than about methodology, graphic design, the technology, etc.

Avoid distorting what the data have to say

Present many numbers in a small space

Make large data sets coherent

Encourage the eye to compare different pieces of data

Reveal the data at several levels of detail

Serve a reasonably clear purpose: description, exploration, tabulation, decoration

Be closely integrated with the statistical and verbal descriptions

Maximize the

“Data-Ink Ratio”

$$\text{ink used to show data} = \frac{\text{data-ink}}{\text{total ink}}$$

1 Erase non-data
ink

2 Erase
redundant
data ink

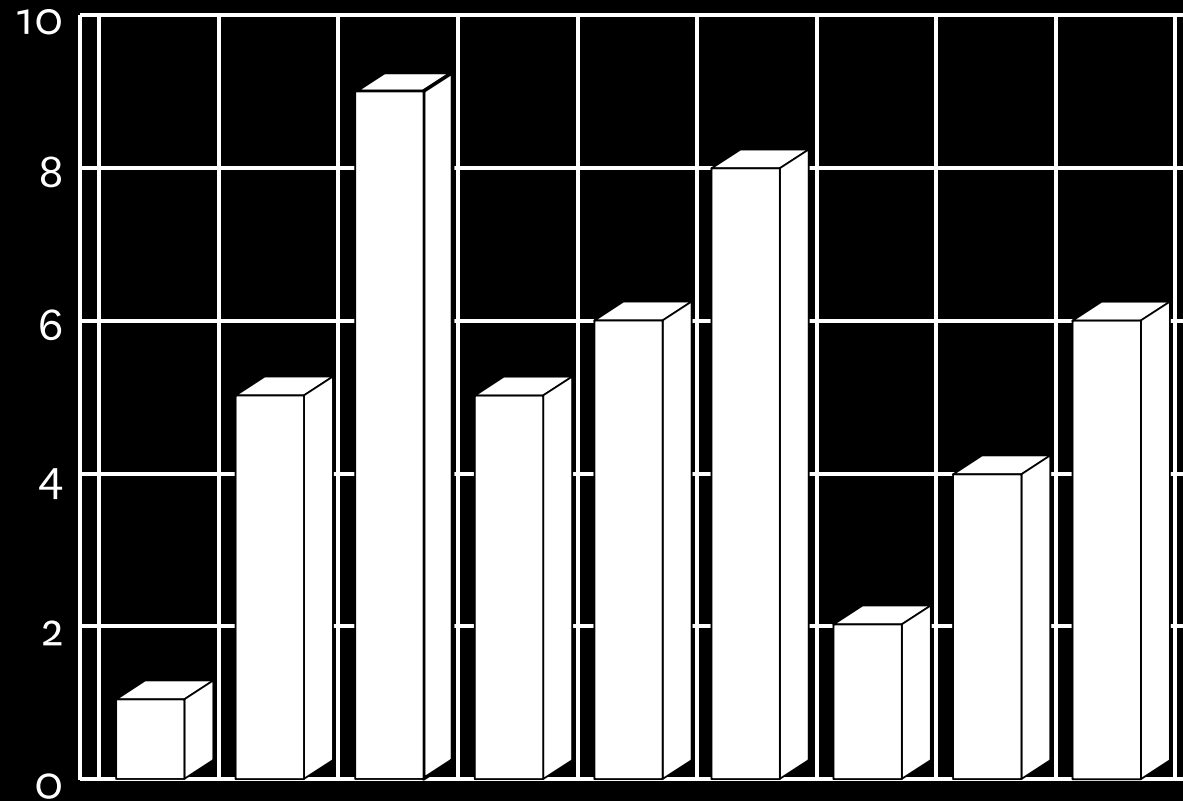
3 Revise and
edit

labeling, annotation, etc.

TIPS FROM TUFTE

Avoid chartjunk

Utilize data-ink

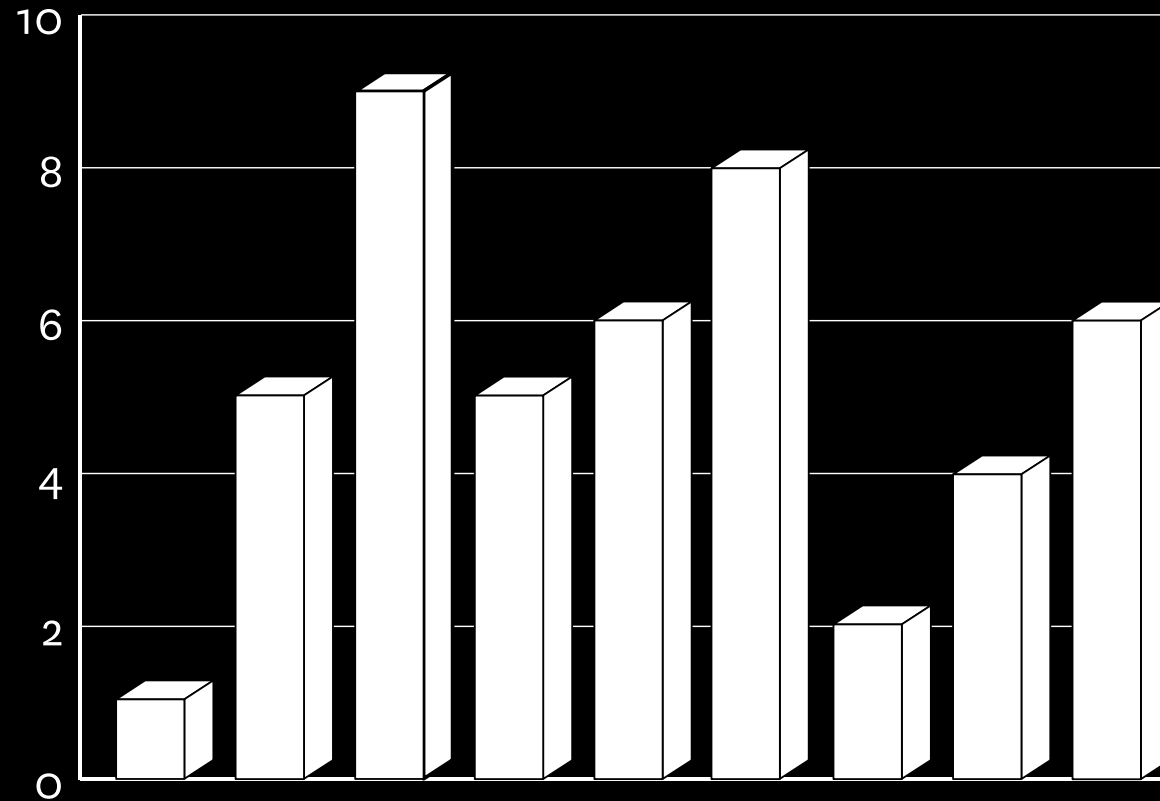


labeling, annotation, etc.

TIPS FROM TUFTE

Avoid chartjunk

Utilize data-ink

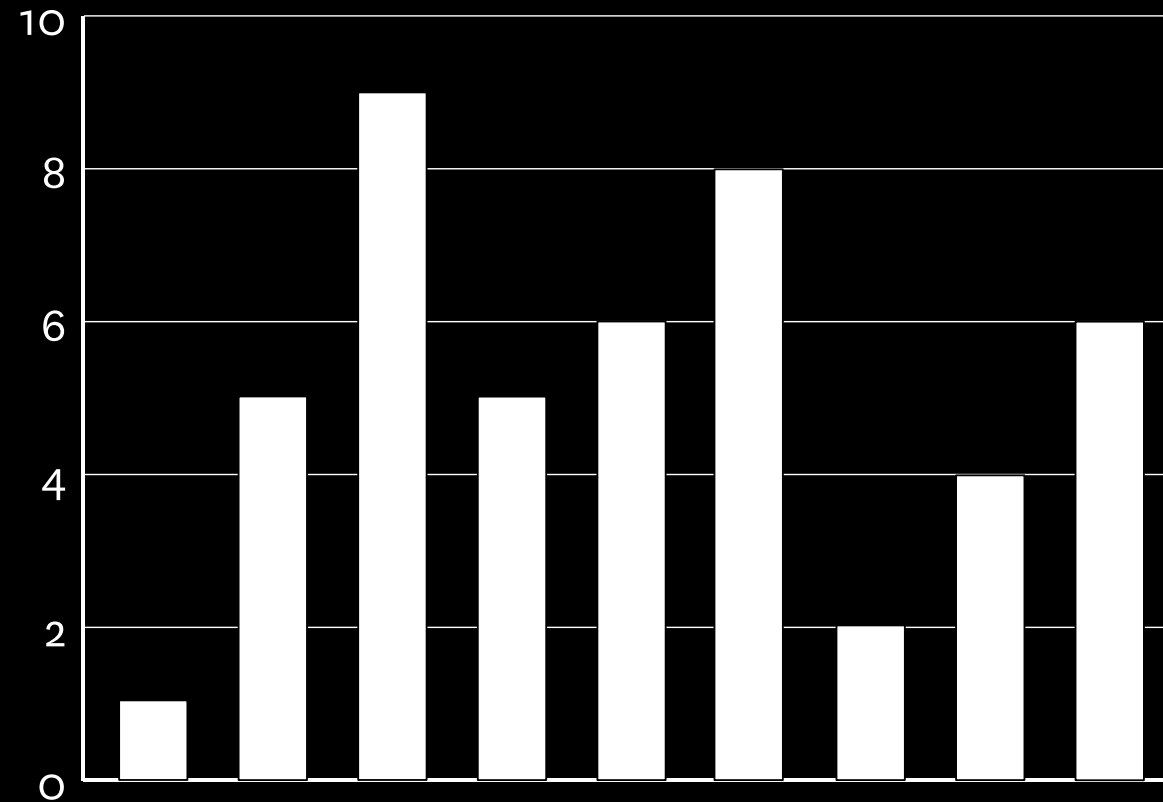


labeling, annotation, etc.

TIPS FROM TUFTE

Avoid chartjunk

Utilize data-ink

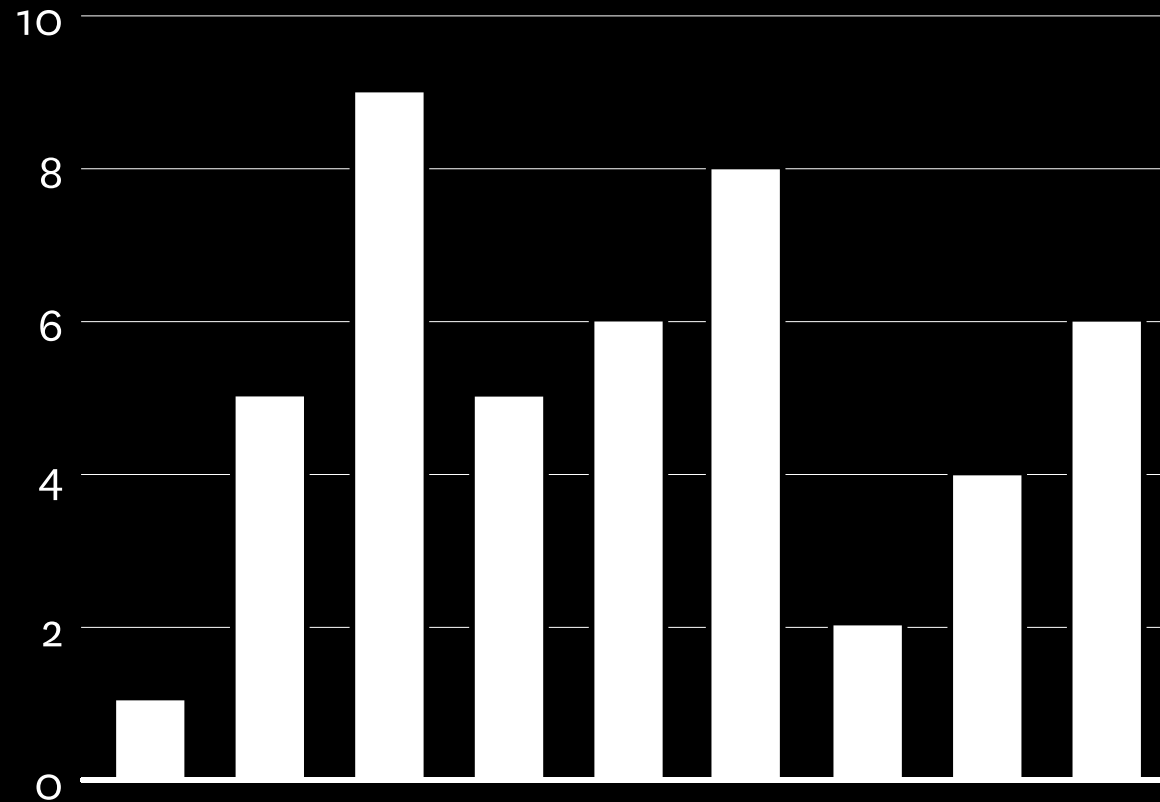


labeling, annotation, etc.

TIPS FROM TUFTE

Avoid chartjunk

Utilize data-ink

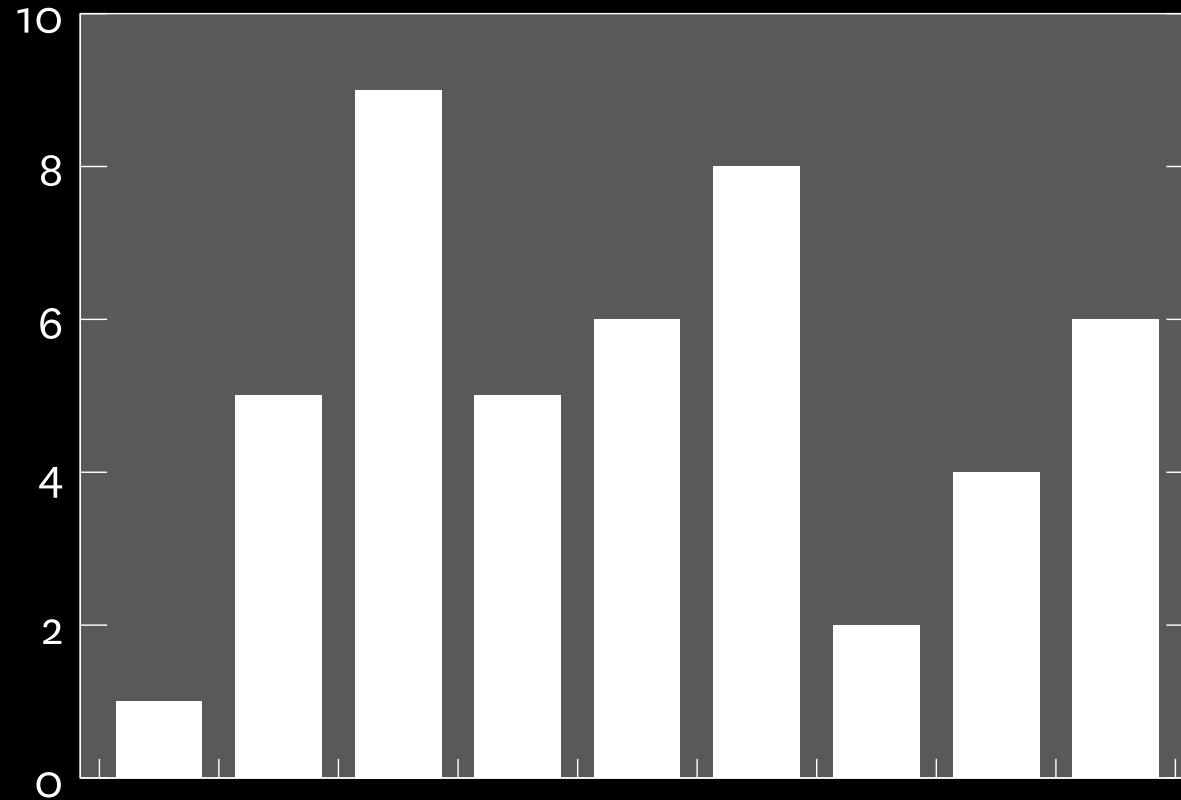


labeling, annotation, etc.

TIPS FROM TUFTE

Avoid chartjunk

Utilize data-ink

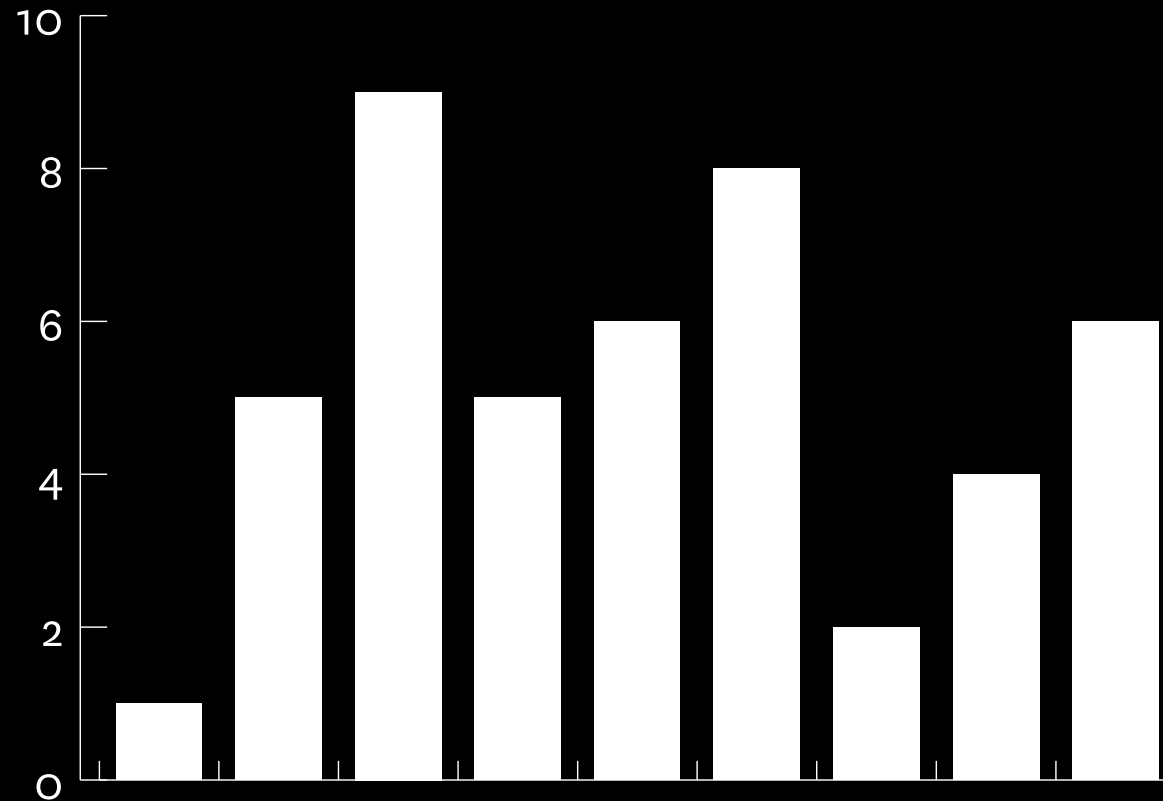


labeling, annotation, etc.

TIPS FROM TUFTE

Avoid chartjunk

Utilize data-ink

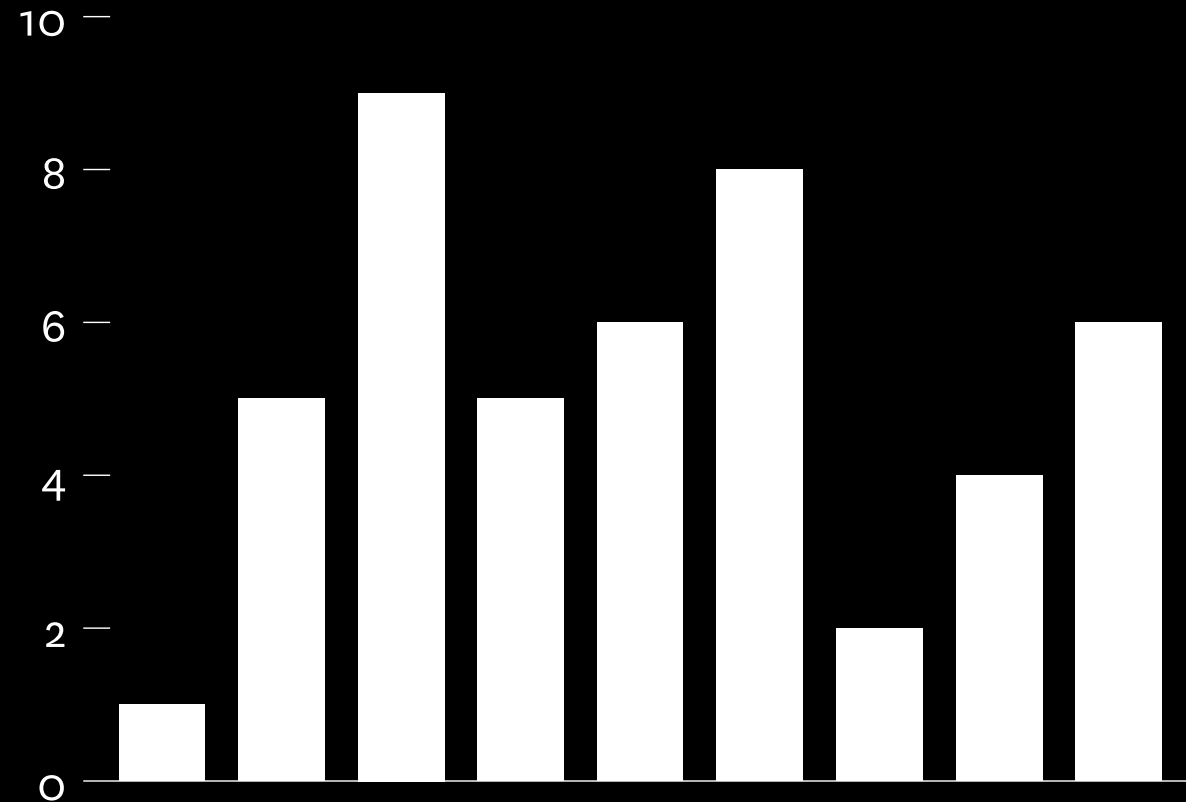


labeling, annotation, etc.

TIPS FROM TUFTE

Avoid chartjunk

Utilize data-ink

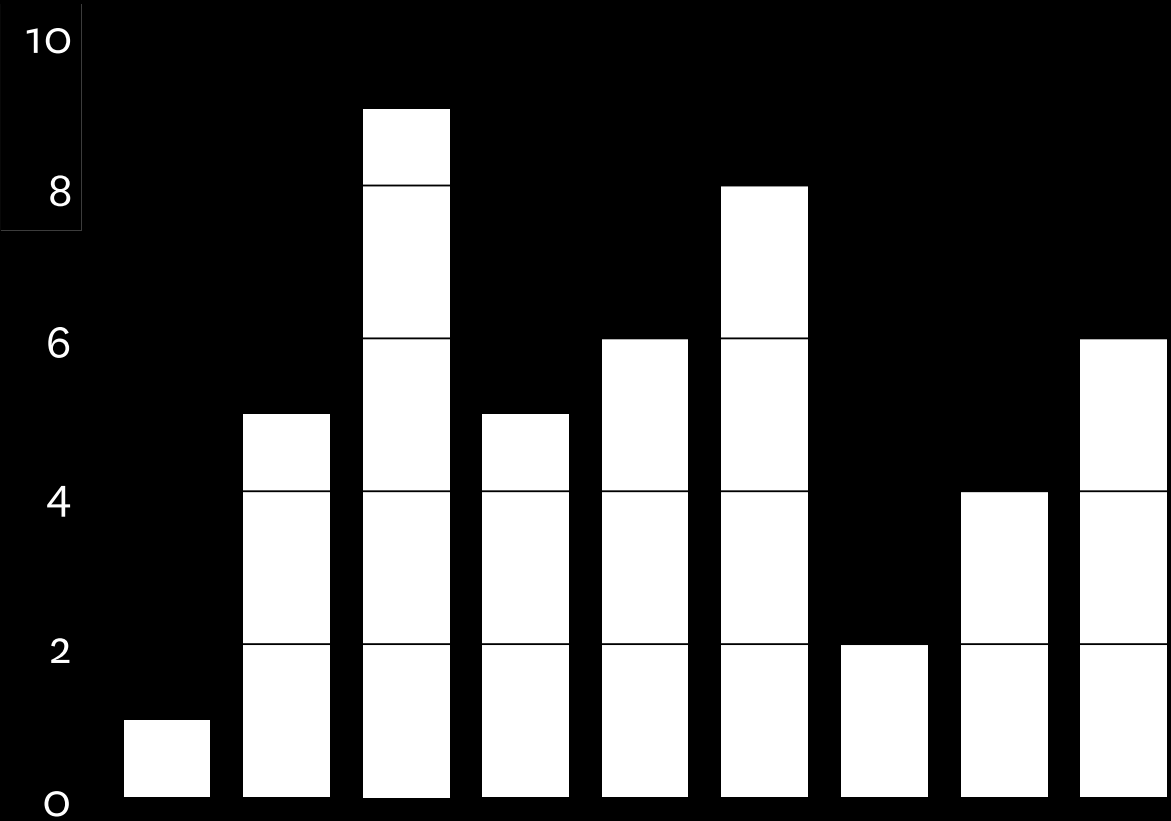


labeling, annotation, etc.

TIPS FROM TUFTE

Avoid chartjunk

Utilize data-ink

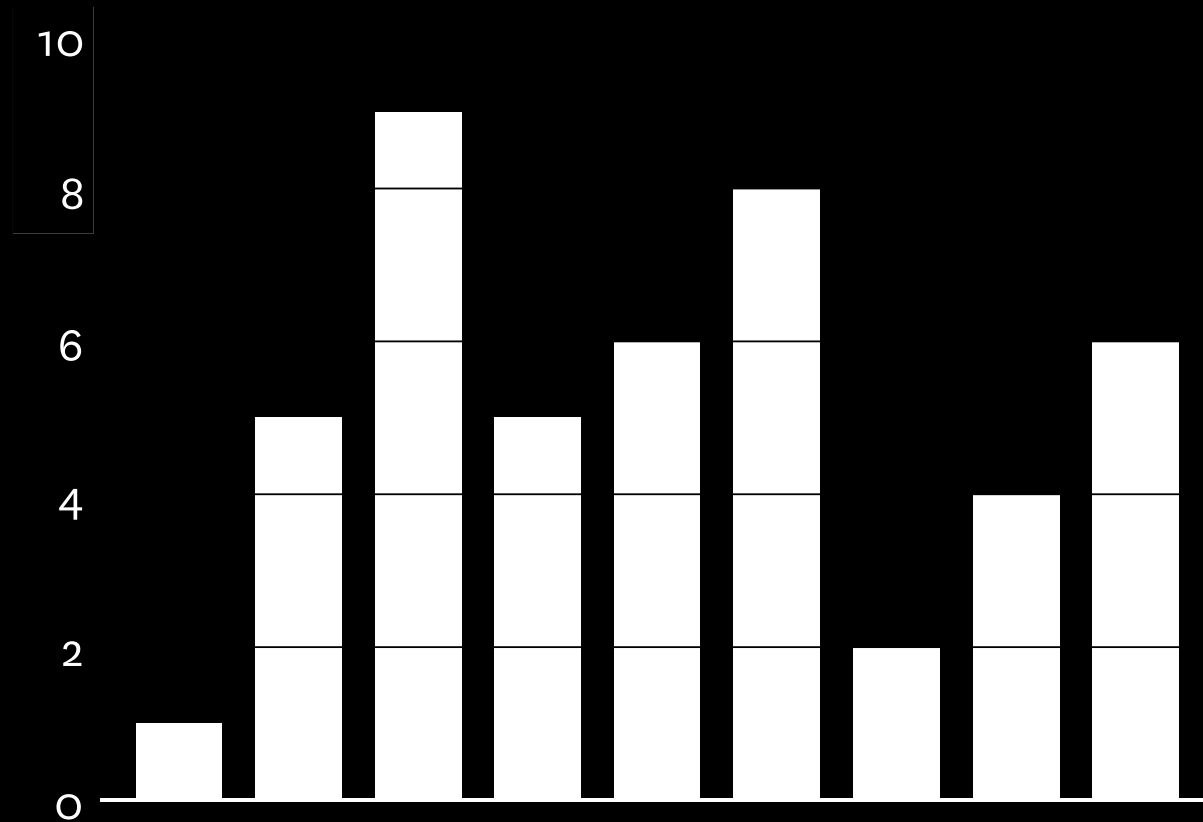


labeling, annotation, etc.

TIPS FROM TUFTE

Avoid chartjunk

Utilize data-ink

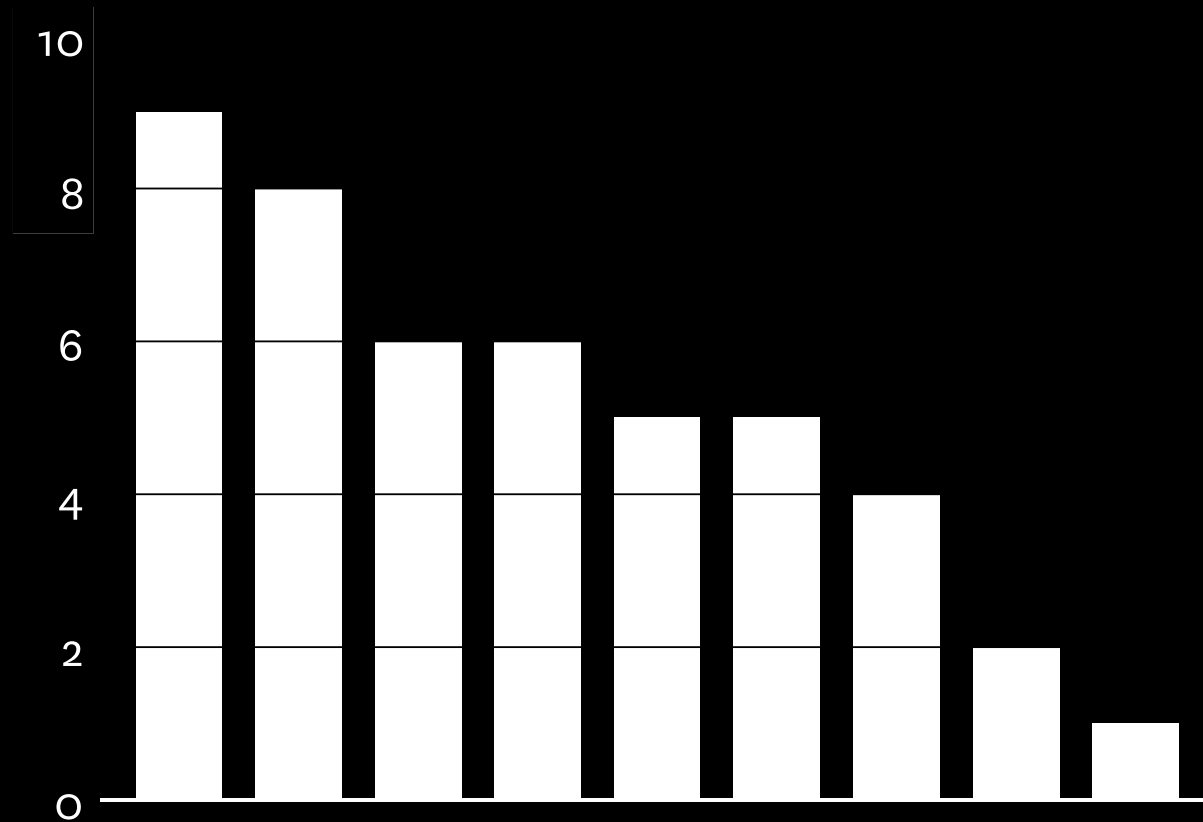


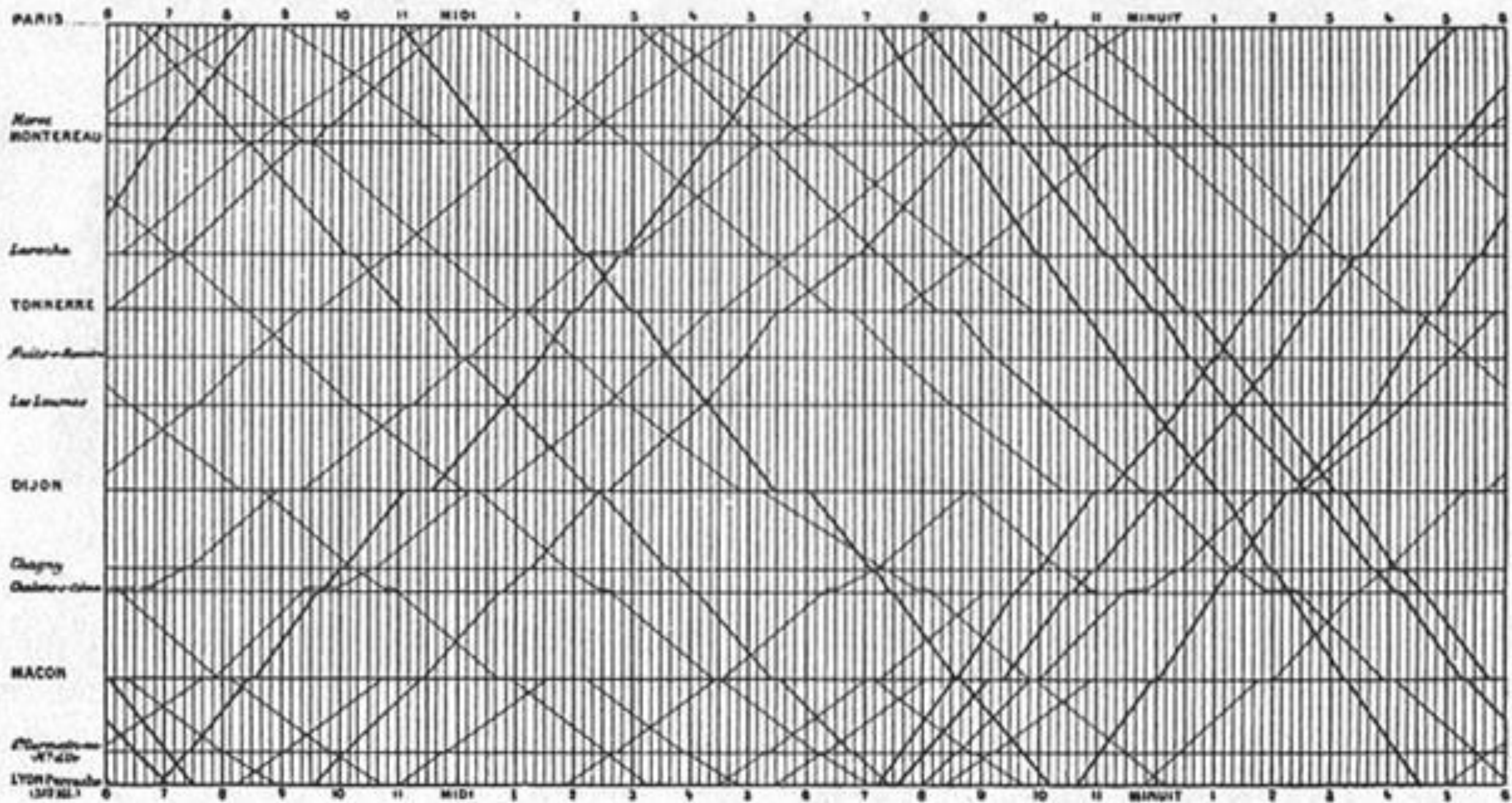
labeling, annotation, etc.

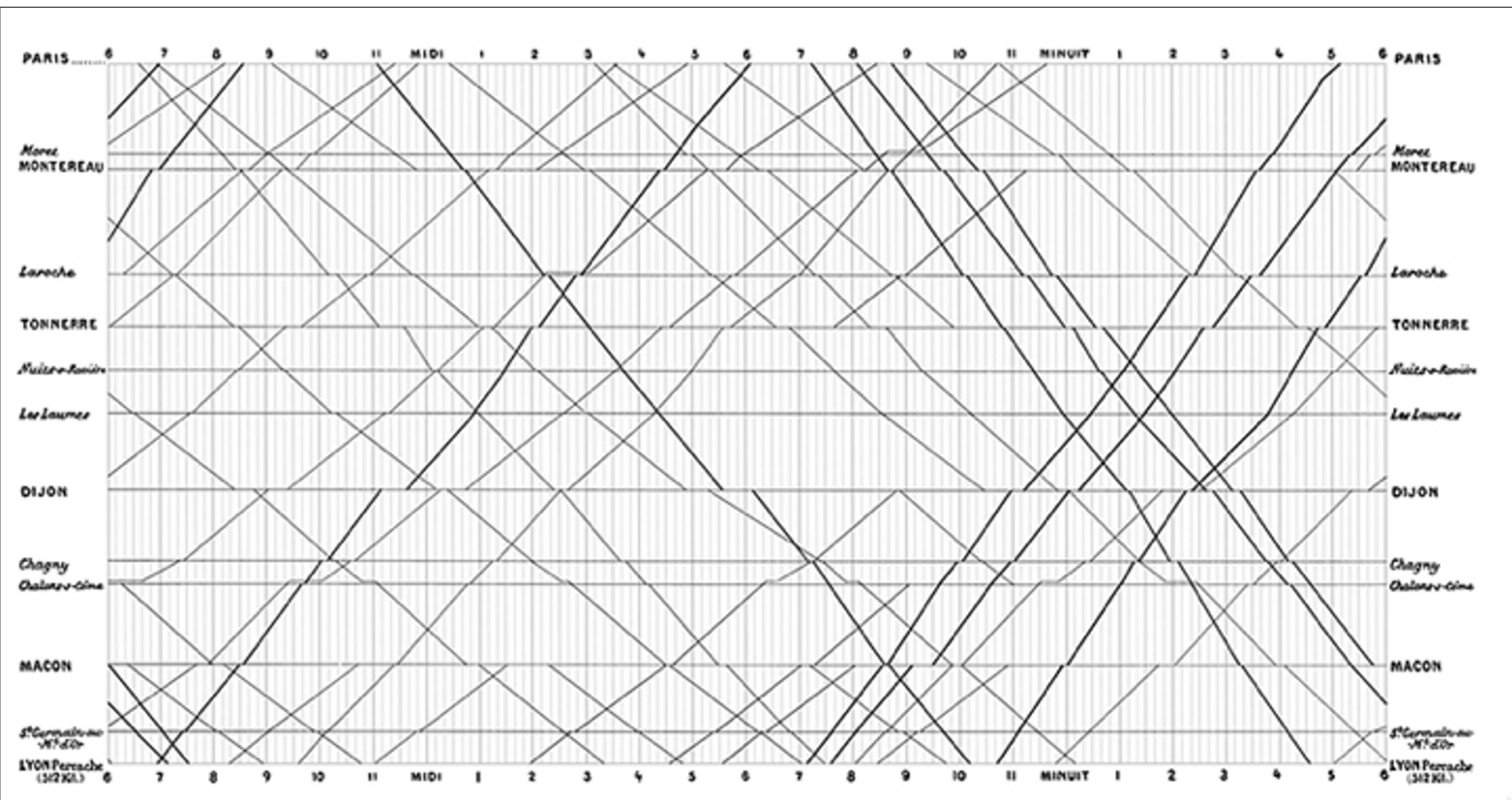
TIPS FROM TUFTE

Avoid chartjunk

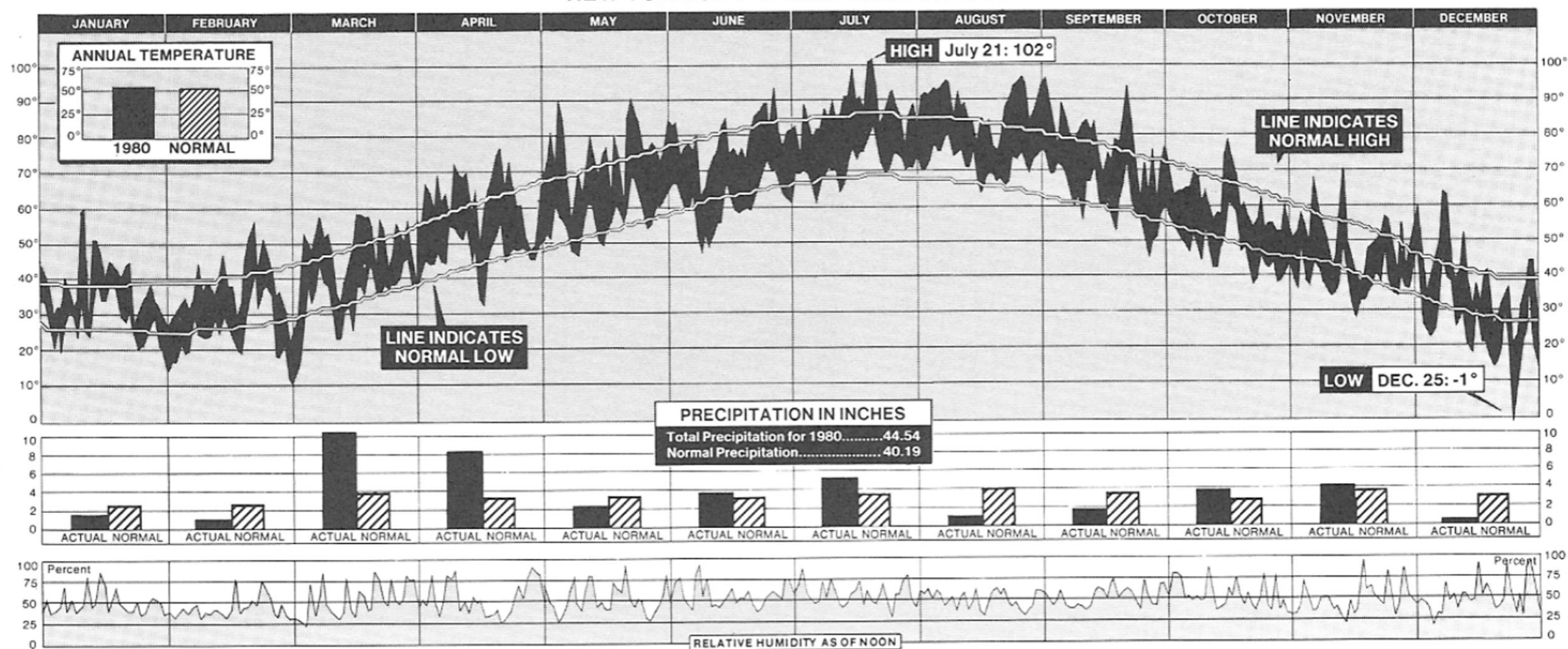
Utilize data-ink





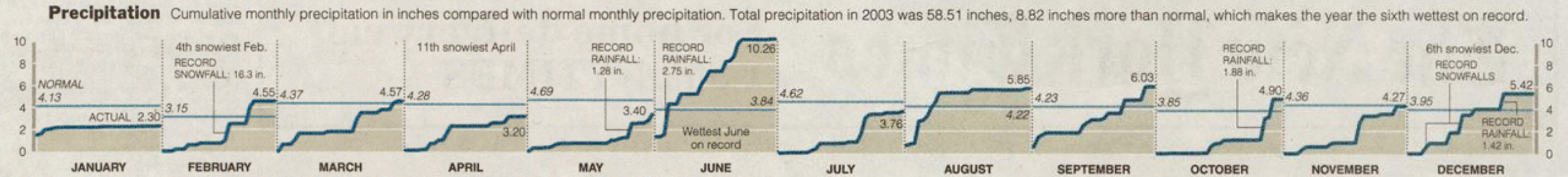
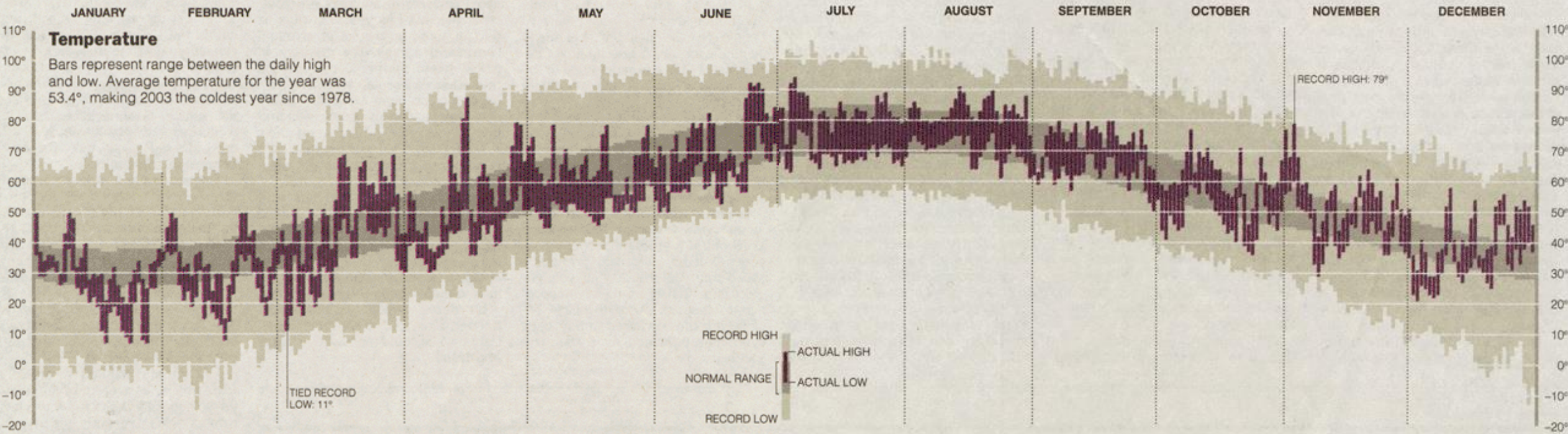


NEW YORK CITY'S WEATHER FOR 1980



New York Times, January 11, 1981, p. 32.

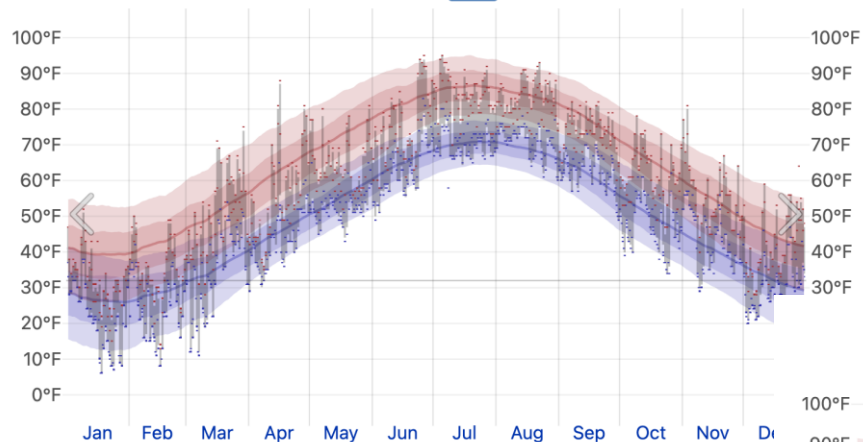
New York City's Weather in 2003



New York City Temperature History 2003

[Link](#) [Download](#) [Compare](#) [Averages](#)

History: 2007 2006 2005 2004 2003 2002 2001 2000 1999

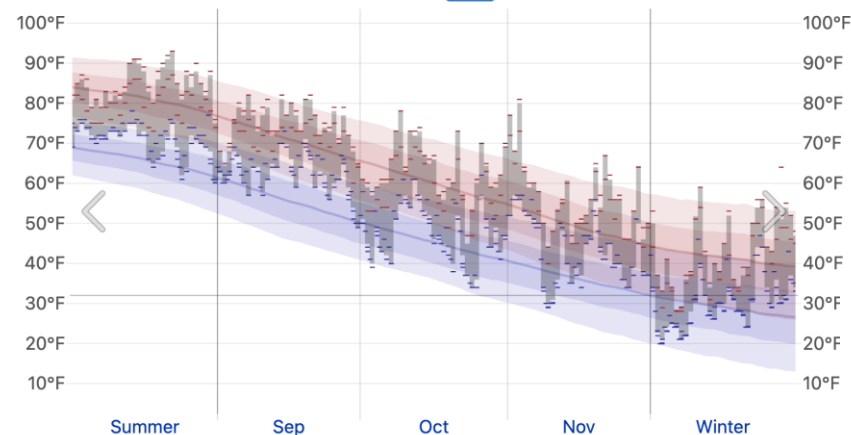


The daily range of reported temperatures (gray bars) and 24-hour highs (red ticks) and lows (blue ticks), placed over the daily average high (faint red line) and low (faint blue line) temperature, with 25th to 75th and 10th to 90th percentile bands.

New York City Temperature History in the Fall of 2003

[← 2003](#) [Link](#) [Download](#) [Compare](#) [Averages](#)

History: 2007 2006 2005 2004 2003 2002 2001 2000 1999

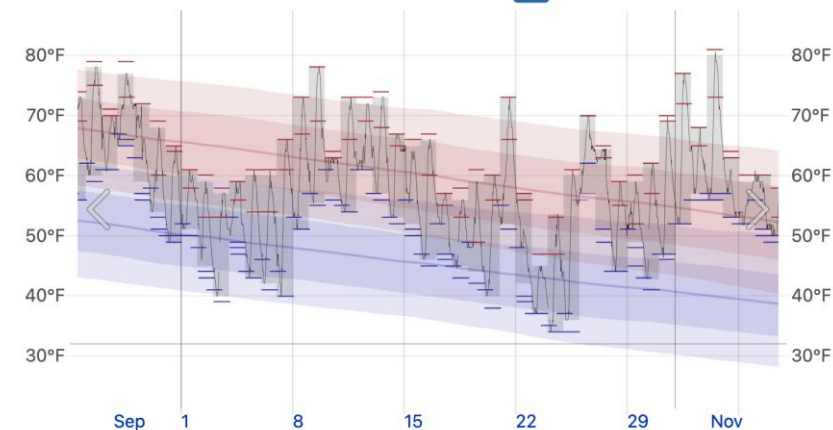


The daily range of reported temperatures (gray bars) and 24-hour highs (red ticks) and lows (blue ticks), placed over the daily average high (faint red line) and low (faint blue line) temperature, with 25th to 75th and 10th to 90th percentile bands.

New York City Temperature History October 2003

[← Fall 2003](#) [Link](#) [Download](#) [Compare](#) [Averages](#)

History: 2005 2004 J F M A M J J A S Oct N D 2002 2001



The daily range of reported temperatures (gray bars) and 24-hour highs (red ticks) and lows (blue ticks), placed over the daily average high (faint red line) and low (faint blue line) temperature, with 25th to 75th and 10th to 90th percentile bands.

Why

What are
our goals?

What questions
do we want to
answer?

Who

Who are
the users?

What are
their tasks?

What

What kind of
data might we
collect?

How might we
convey the
information
associated with
this data?

Graphical
Integrity

Accurate

**Presentation size
should precisely
match data**

**Avoid 2D (area) and
3D (volume) for 1D
data**

Clear

**Label carefully
and clearly**

In Context

**Present data in
context**

**Adjust currency
values for inflation or
other correlated
changes (population
changes, etc)**

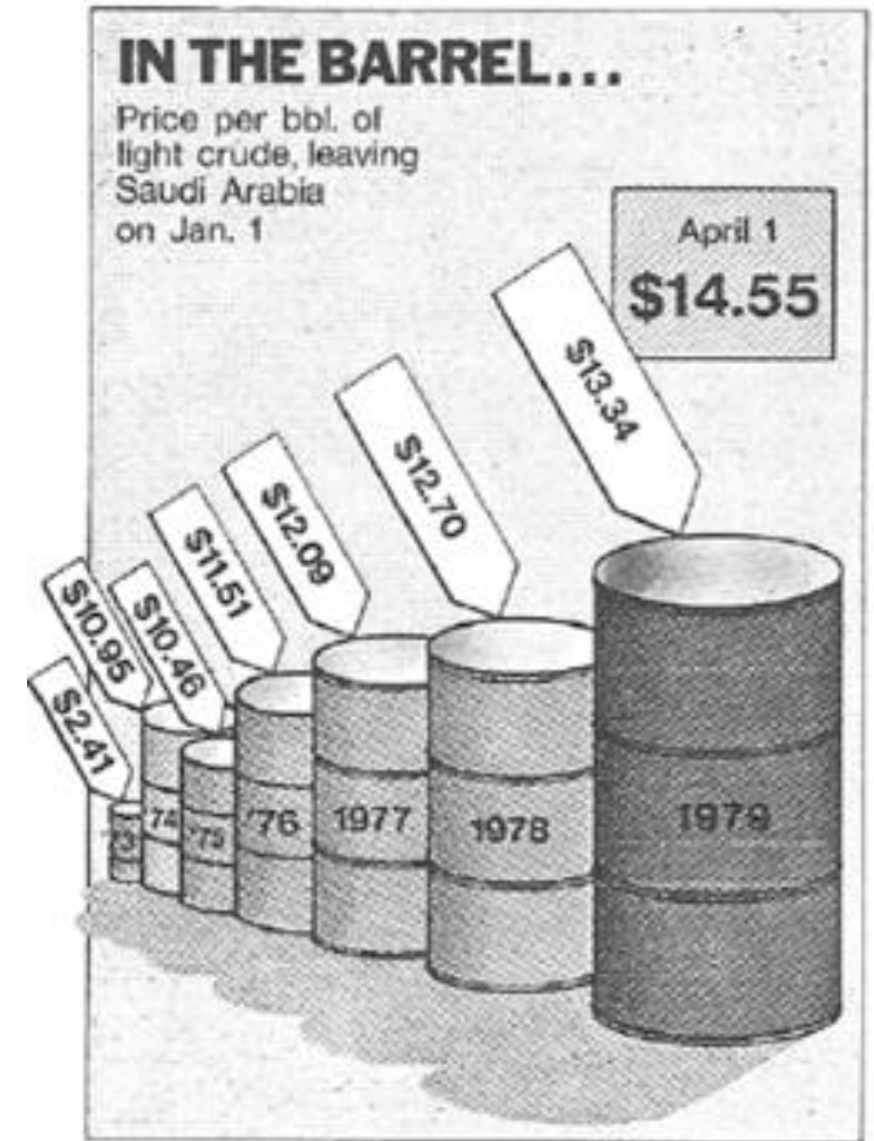
lie factor = $\frac{\text{size of effect in graphic}}{\text{size of effect in data}}$

Truth = 0.95 to 1.05 (5% error)

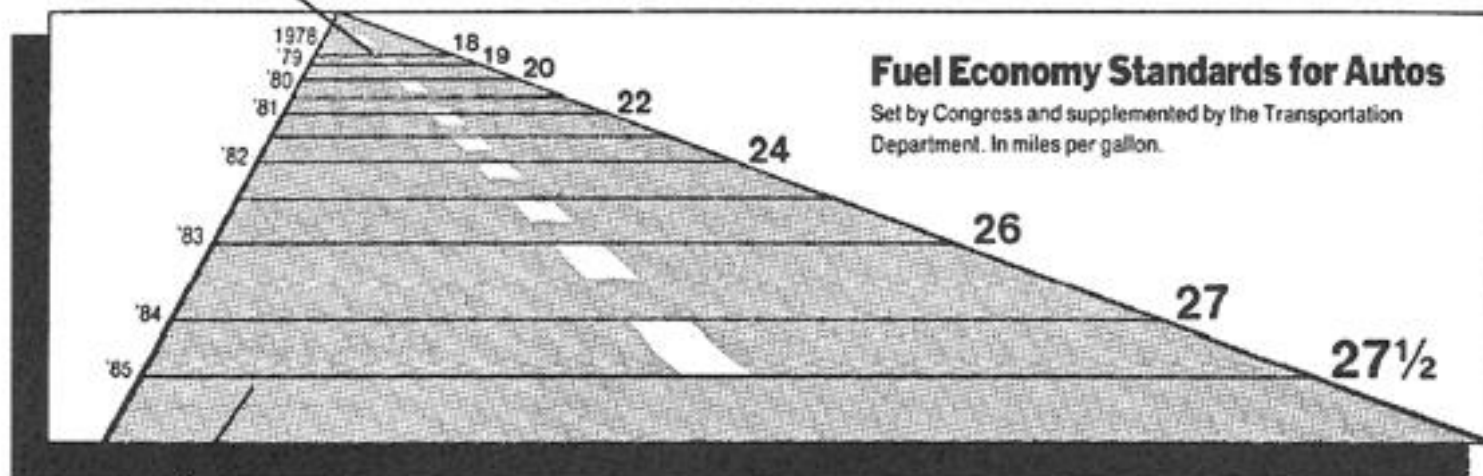
2 to 5 is not uncommon.

Lie factor by area = 9.4

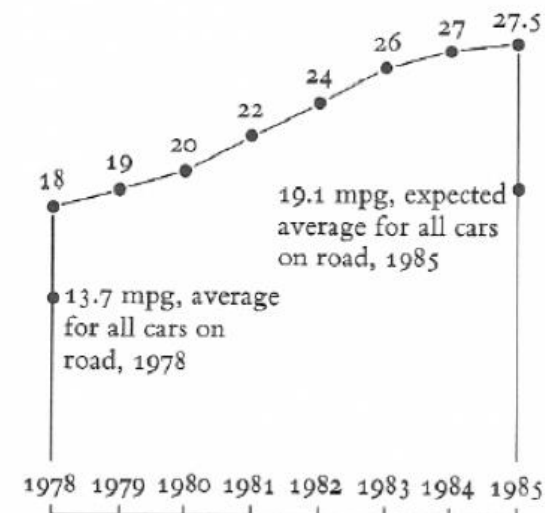
Lie factor by volume = 59.4



This line, representing 18 miles per gallon in 1978, is 0.6 inches long.



This line, representing 27.5 miles per gallon in 1985, is 5.3 inches long.



Required Fuel Economy Standards:
New cars built from 1978 to 1985

New York Times, August 9, 1978, p. D-2.

Lie factor: 14.8

Gratuitous 3D and
perspective

Non-uniform intervals and
scales

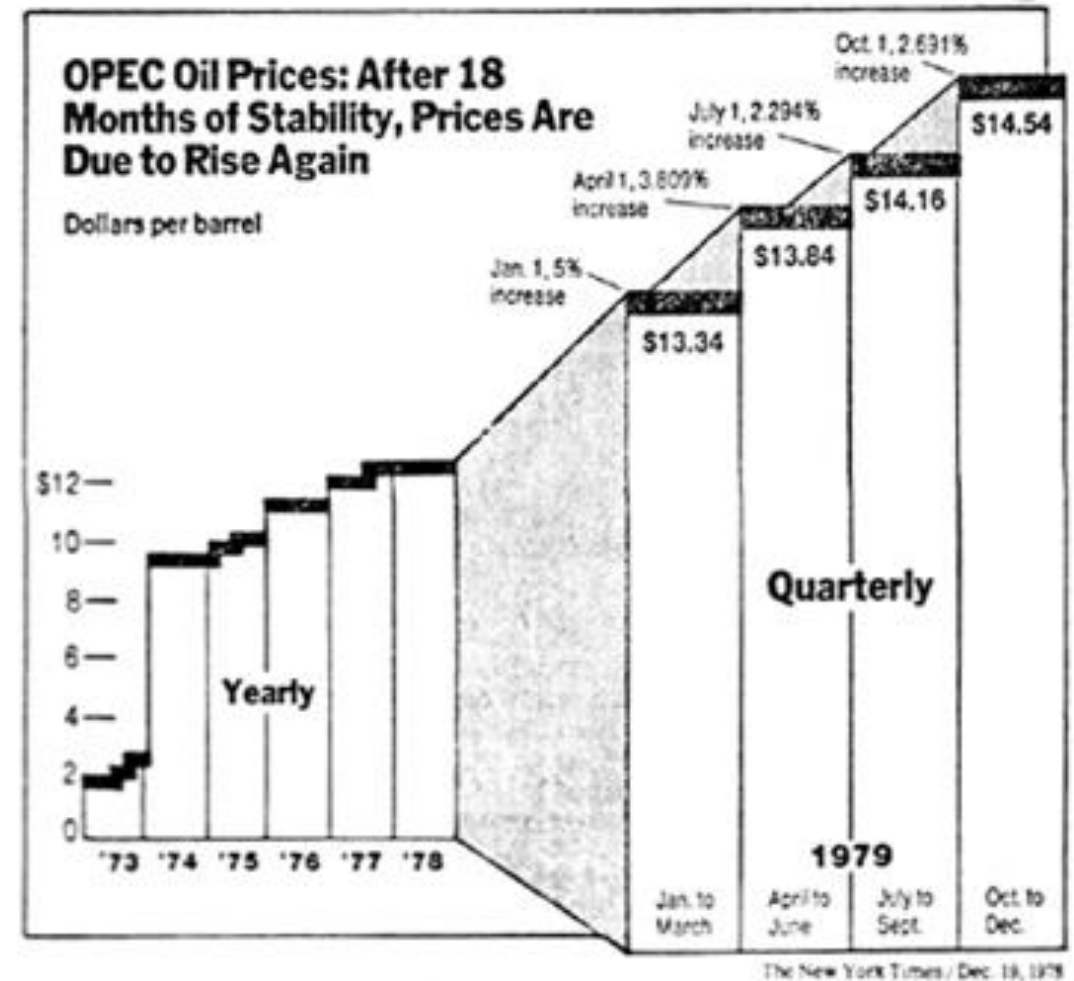
Design distortion:

Yearly vs. Quarterly

Nonlinear scale

Perspective distortion

Annual increase is actually
similar to many previous
years!



New York State
Total Budget Expenditures and
Aid to Localities

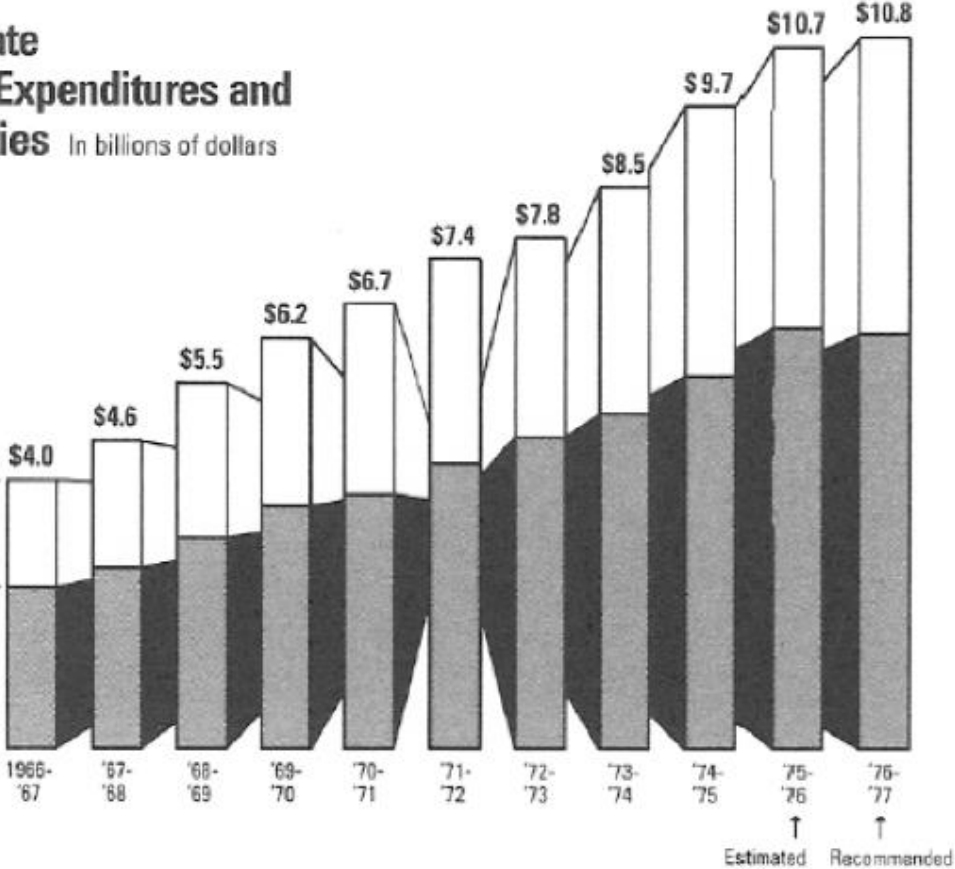
In billions of dollars

Fiscal 1966-1976

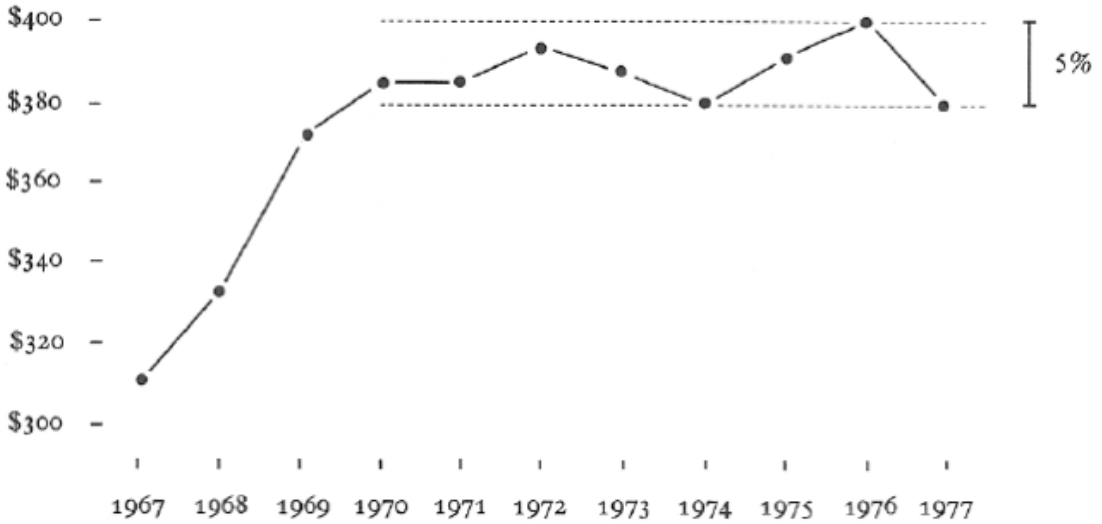
Total Budget

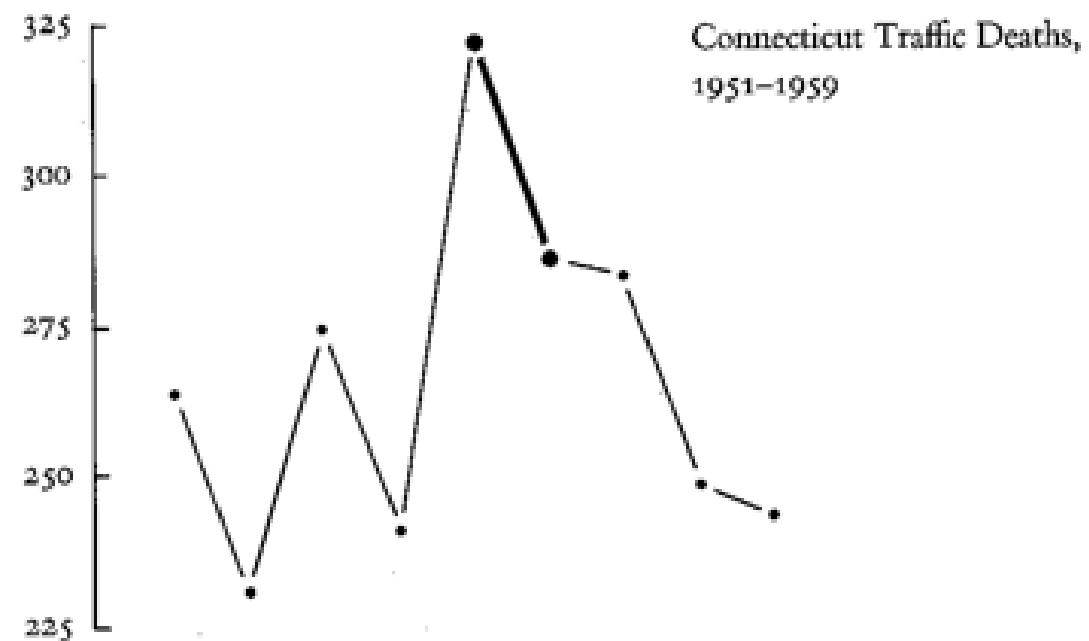
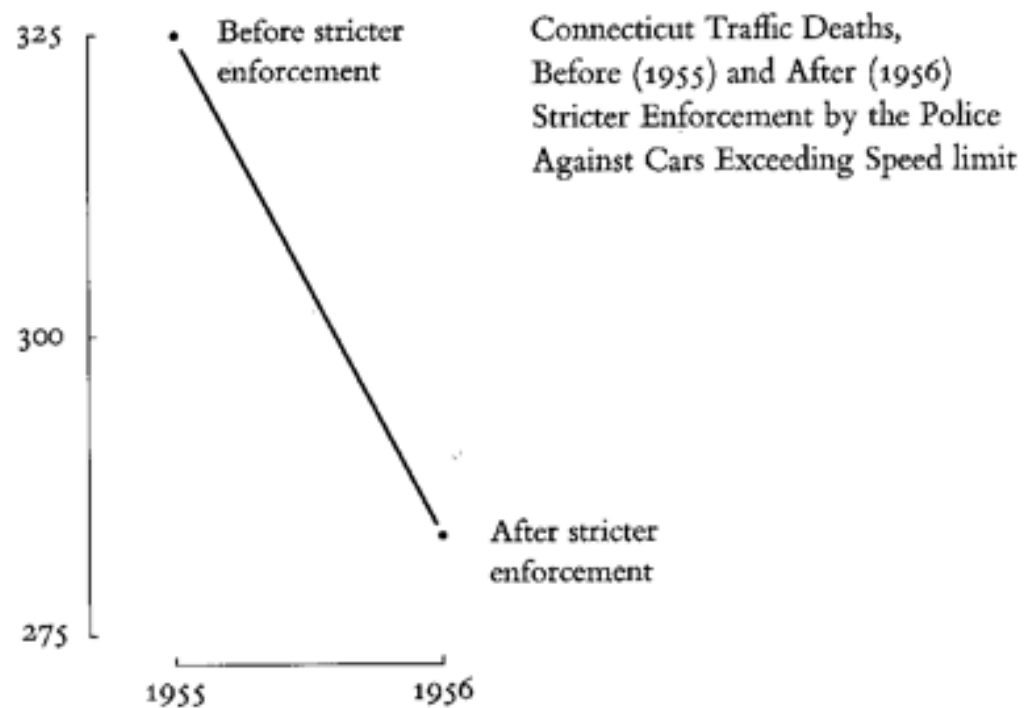
Total Aid to
Localities*

*Varying from a low
of 56.7 percent of
the total in 1970-71
to a high of 60.7
percent in 1972-73



Per capita
budget expenditures,
in constant dollars





Greatest number of ideas

In the shortest time

With least ink

In the smallest space

**Graphical
Excellence**

**Present value
relationships accurately**

Size matches data

Avoid area and volume encodings

**Adjust currency values for inflation,
etc.**

**Label carefully and
clearly**

Present data in context

**Graphical
Integrity**

**Clear, detailed,
complete**

Provides precise values

**Indicates important
events**

Defines units, context

**Corrects for presentation
distortion**

**Label directly
on the graphic**

**Makes graphic self-
explanatory**

Maintains focus

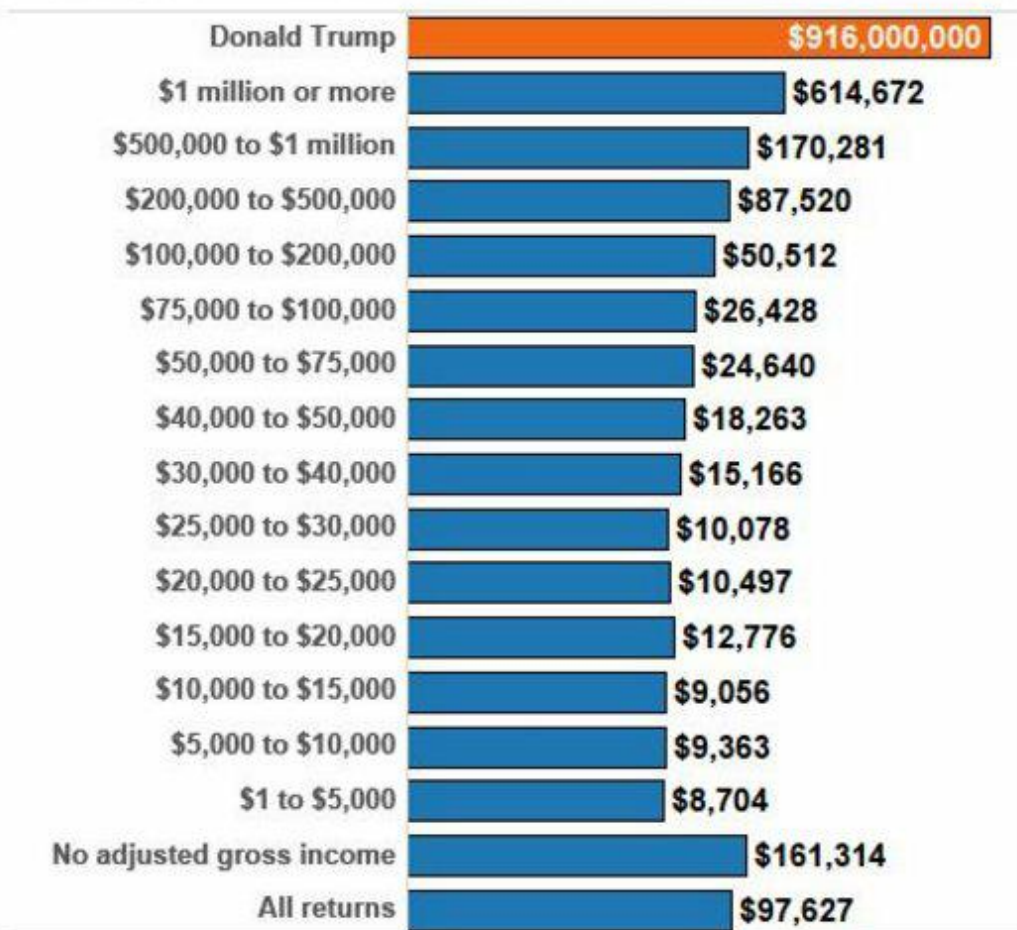
~~Legends~~

Trump's tax loss whopper



About a half million individual filers offset their taxable income with net operating losses in 1995, but the average amounts didn't come close to the tax break Donald Trump claimed that year. Here are the average amounts of net operating losses claimed by taxpayers based on levels of adjusted gross income. SOURCE: IRS SOI, 1995 data

Adjusted gross income | Average net operating loss claimed



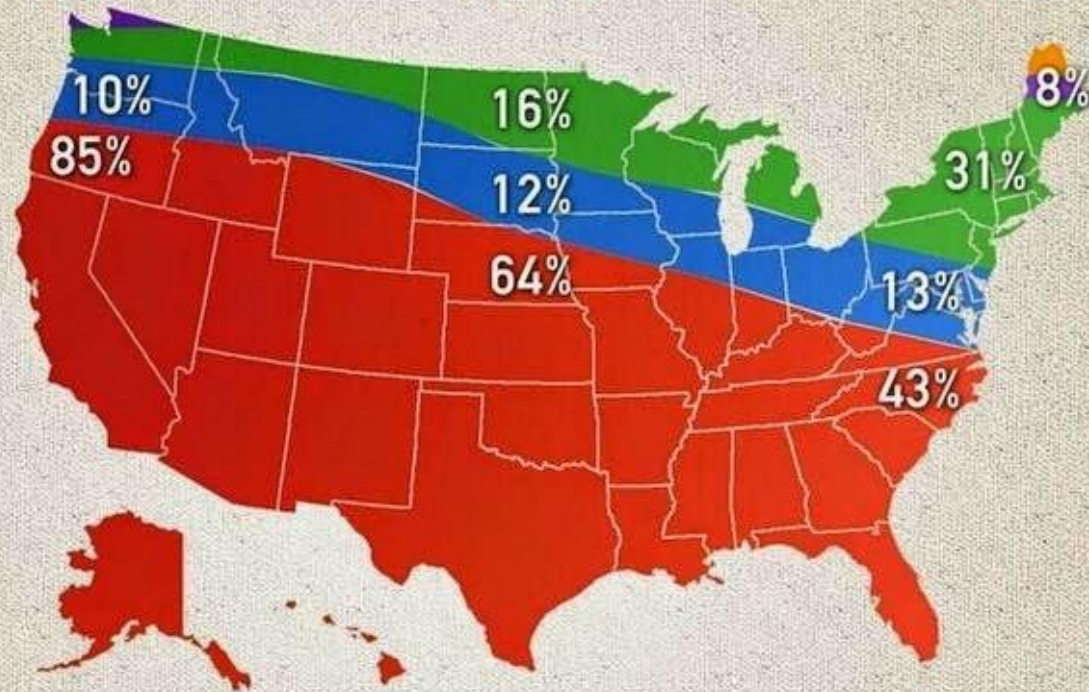
CHANGING FACE OF AMERICA

Percent of total U.S. population by race and ethnicity, 1960-2060

1960

2010

2060



OTHER

ASIAN

HISPANIC

BLACK

WHITE

SOURCE: PEW RESEARCH CENTER

 **NIGHTLY NEWS**
with BRIAN WILLIAMS

Critiquing Visualizations

1

Consider the purpose of the visualization and the intended audience.

2

Ascertain your initial reaction.

3

Examine the visualization in detail.

Then, answer questions like the following...

- 1** Is the design visually appealing/aesthetically pleasing?
- 2** Is it immediately understandable? If not, is it understandable after a short period of study?
- 3** Does it provide insight or understanding that was not obtainable with the original representation (text, table, etc.)?
- 4** Does it provide insight or understanding better than some alternative visualization would? Or does it require excessive cognitive effort?
- 5** What kind of visualization might have been better?
- 6** Does the visualization reveal trends, patterns, gaps, and/or outliers? Can the viewer make effective comparisons?
- 7** Does the visualization successfully highlight important information, while providing context for that information?
- 8** Does it distort the information? If it transforms it in some way, is this misleading or helpfully simplifying?
- 9** Does it omit important information?
- 10** Is it memorable?
- 11** Does it use visual components properly? That is, does it properly represent the data using lines, color, position, etc.? Does it transform nominal, ordinal, and quantitative information properly?
- 12** Does it use labels and legends appropriately?

Activity

In groups of 2:

- 1 Find a visualization (good, bad or ugly)**
- 2 Jot down answers to the Visualization Evaluation Checklist**
- 3 Each person sketch a solution to an issue you identified in the visualization**
- 4 Trade critique for feedback**
- 5 Submit to Canvas**

Break

Until 9:25pm

Project Groups

1

**Liz Overbeek
Raquel Eisele
Julie Wang
Kavya Nagalakunta**

2

**Anoushka Halder
Tasnim Hasan
Ambika Ramesh Menon
Shreya Saxena**

3

**Jessica Hsu
Ralph Chang
Celia Tsao Trushaa
Ramanan**

4

**Amy La
MJ Schneider
Jacqueline Yan (Jackie)
Catherine Pan**

5

**Gily Yu
Sonali Arbune
Yiting Wang
Mai Kanapornchai**

6

**Ruiqi Chen
Amy Lee
Samantha Xiao
Honglam Van**

7

**Dina Chawla
Ko Motonaga
Jeremy Inducil
Julie Lee**

Names are taken from student survey

Canvas Groups to be populated soon

Assignment

Select **one static visualization** from any source (online or elsewhere)

Use the methods discussed in class to **critique it** (800-1000 words, plus screenshots)

Follow **guidance** in Canvas assignment

Your critique will be anonymously **peer-reviewed** and graded

Instructions about peer-review will be given next week

Peer review **grades** will be **confirmed** by Donhoon & Doug

What do you want to design and build?

What is the **purpose** of the visualization?

What **results** do you anticipate?

Goal is to get **feedback** on the scope

Talk to intended **users** about their needs!

What **data** you might use (more next week)

Early **concepts** and sketches

Project **logistics**

Tools

Project Proposal